

INTERNSHIP FINAL REPORT

Bluestock Mutual Fund Analytics Capstone Project

*A Comprehensive Study on Mutual Fund Data Analytics and Investment
Decision Support*

STUDENT NAME

ROHIT J

ORGANIZATION

Bluestock Internship Program

SUBMISSION DATE

June 2026

Acknowledgement

The successful completion of this capstone project would not have been possible without the guidance, support, and encouragement of several individuals and institutions. I take this opportunity to express my sincere gratitude to all those who contributed to this endeavour.

First and foremost, I extend my deepest appreciation to the **Bluestock Internship Program** for providing me with this invaluable opportunity to work on a real-world financial analytics project. The structured learning environment and exposure to industry-standard tools and methodologies have significantly enhanced my understanding of mutual fund analytics and data-driven investment decision-making.

I am profoundly grateful to my mentors and supervisors at Bluestock, whose expert guidance, constructive feedback, and continuous support throughout the internship period were instrumental in shaping the direction and quality of this project. Their willingness to share knowledge and provide insights into the nuances of financial data analytics has been truly enlightening.

I would also like to acknowledge the faculty members of my academic institution for their foundational teachings in statistics, data analytics, and financial management. The theoretical knowledge imparted during my coursework provided the necessary framework for this practical application.

Special thanks are extended to my peers and colleagues who engaged in meaningful discussions, offered alternative perspectives, and provided moral support during challenging phases of the project. Their collaborative spirit and encouragement were invaluable.

Finally, I express my heartfelt gratitude to my family for their unwavering support, patience, and understanding throughout this journey. Their belief in my abilities served as a constant source of motivation.

ROHIT J

June 2026

Executive Summary

The Indian mutual fund industry has witnessed unprecedented growth over the past decade, with Assets Under Management (AUM) crossing significant milestones and investor participation reaching record levels. However, the vast array of available schemes, coupled with the complexity of financial metrics, presents considerable challenges for both novice and experienced investors in making informed investment decisions. This capstone project addresses these challenges by developing a comprehensive mutual fund analytics framework that integrates data engineering, statistical analysis, and interactive visualisation techniques.

The primary purpose of this project was to design and implement an end-to-end analytics solution capable of processing, analysing, and visualising mutual fund data to derive actionable investment insights. The project leveraged a robust technology stack comprising **Python** as the primary programming language, **Pandas** and **NumPy** for data manipulation and numerical computations, **Jupyter Notebook** as the interactive development environment, and **Power BI** for dashboard creation and business intelligence reporting.

The analytical framework encompassed multiple dimensions of mutual fund evaluation. An Extract-Transform-Load (ETL) pipeline was designed to standardise and integrate data from nine distinct datasets covering fund master information, Net Asset Value (NAV) history, investor transactions, scheme performance, SIP inflows, category flows, portfolio holdings, benchmark indices, and industry folio counts. Exploratory Data Analysis (EDA) was conducted to understand investor demographics, industry trends, fund performance distributions, risk profiles, and expense structures. Advanced financial metrics, including the Sharpe Ratio, Alpha, Beta, Sortino Ratio, Historical Value-at-Risk (VaR), and Conditional Value-at-Risk (CVaR), were computed to evaluate risk-adjusted performance. Additionally, investor cohort analysis, SIP continuity analysis, a mutual fund recommendation system based on risk appetite, and portfolio concentration assessment using the Herfindahl-Hirschman Index (HHI) were implemented to provide comprehensive decision support.

The business value of this project lies in its ability to transform raw, disparate mutual fund data into structured, meaningful insights that can guide investment decisions. The interactive Power BI dashboards developed as part of this project enable stakeholders to explore data dynamically, identify trends, and compare fund performance across multiple dimensions. The recommendation system offers personalised fund suggestions based on individual risk profiles, while the risk metrics provide quantitative measures of potential downside exposure.

The outcomes of this project include a fully operational ETL pipeline, a comprehensive set of analytical reports with nineteen data visualisations, four interactive Power BI dashboards, and a documented framework for mutual fund evaluation. All project objectives were successfully achieved, demonstrating the practical application of data analytics in financial services and the potential for data-driven approaches to enhance investment decision-making processes.

Table of Contents

Acknowledgement

Executive Summary

1. Problem Statement

2. Project Objectives

3. Data Sources

4. ETL Design and Data Preparation

4.1 Extract Phase

4.2 Transform Phase

4.3 Load Phase

5. Exploratory Data Analysis

5.1 Investor Demographics

5.2 Mutual Fund Industry Trends

5.3 Fund Performance Analysis

5.4 Risk and Expense Analysis

5.5 NAV and Portfolio Analysis

5.6 SIP Behaviour Analysis

5.7 Summary of EDA Findings

6. Performance Analysis

7. Advanced Analytics and Risk Metrics

7.1 Historical Value at Risk (VaR)

7.2 Conditional Value at Risk (CVaR)

7.3 Rolling 90-Day Sharpe Ratio

7.4 Investor Cohort Analysis

7.5 SIP Continuity Analysis

7.6 Mutual Fund Recommendation System

7.7 Sector HHI Concentration Analysis

8. Power BI Dashboard Development

9. Key Findings

10. Limitations

11. Recommendations

12. Future Scope

13. Conclusion

1. Problem Statement

The Indian mutual fund industry, regulated by the Securities and Exchange Board of India (SEBI) and represented by the Association of Mutual Funds in India (AMFI), has emerged as one of the most dynamic segments of the country's financial services sector. With over forty Asset Management Companies (AMCs) offering thousands of schemes across diverse categories, including equity, debt, hybrid, solution-oriented, and other funds, investors are confronted with an overwhelming array of choices. This abundance of options, while indicative of a mature and competitive marketplace, creates significant challenges in scheme selection and portfolio construction.

Individual investors, particularly those with limited financial literacy, frequently struggle to navigate the complex landscape of mutual fund offerings. The decision-making process involves evaluating numerous quantitative and qualitative factors, including historical returns, risk measures, expense ratios, fund manager track records, portfolio composition, and alignment with personal financial goals. Without systematic analytical tools, investors often rely on anecdotal evidence, promotional materials, or simplified star ratings that may not capture the full risk-return profile of a given scheme.

Traditional evaluation methods present several limitations that hinder effective decision-making. Many investors depend solely on point-to-point returns, which fail to account for the volatility experienced during the investment period and the consistency of performance across different market cycles. Standard return metrics do not adjust for risk, making it impossible to determine whether superior returns were achieved through skilled management or simply by assuming greater risk. Furthermore, the lack of integration across different data dimensions, such as combining investor behaviour patterns with fund performance characteristics, prevents the development of holistic investment insights.

The importance of integrated analytics for investment decision-making cannot be overstated in the current financial environment. As market dynamics become increasingly complex and interconnected, the ability to synthesise information from multiple sources and analytical perspectives becomes a critical competitive advantage. Integrated analytics enables the identification of patterns and relationships that would remain invisible in isolated analyses. For instance, understanding how SIP continuity rates correlate with market volatility and investor demographics can inform both product design and investor education initiatives. Similarly, combining risk metrics with portfolio concentration measures provides a more comprehensive view of potential downside exposure than either analysis alone.

This project addresses these challenges by developing a comprehensive analytics framework that integrates multiple data sources, applies rigorous statistical and financial methodologies, and presents findings through interactive visualisations. The solution bridges the gap between raw data availability and actionable investment intelligence, providing stakeholders with the analytical tools necessary to make informed, evidence-based mutual fund investment decisions.

2. Project Objectives

The Bluestock Mutual Fund Analytics Capstone Project was designed with nine specific, measurable objectives that collectively address the challenges identified in the problem statement. Each objective was

formulated to contribute a distinct analytical capability to the overall framework, ensuring comprehensive coverage of mutual fund data processing, analysis, and visualisation requirements.

The first objective focused on the design and implementation of an **ETL pipeline for mutual fund datasets**. This foundational component was essential for ensuring data quality, consistency, and accessibility across all subsequent analytical operations. The ETL pipeline was required to handle multiple disparate data sources, each with its own structure, format, and update frequency, and to produce a unified, analysis-ready dataset repository.

The second objective centred on **exploratory data analysis** to understand the characteristics, distributions, and relationships present in the mutual fund data. This phase was critical for identifying data quality issues, discovering patterns and anomalies, and generating hypotheses for deeper investigation. EDA served as the analytical foundation upon which all subsequent modelling and evaluation activities were built.

The third objective involved **mutual fund performance evaluation using established financial metrics**. This included the computation of risk-adjusted performance measures such as the Sharpe Ratio, Alpha, Beta, and Sortino Ratio, which provide standardised frameworks for comparing funds with different risk profiles and investment strategies.

The fourth and fifth objectives addressed risk measurement through the computation of **Historical Value at Risk (VaR) and Conditional Value at Risk (CVaR)** at the 95% confidence level. These metrics quantify the potential magnitude of losses under adverse market conditions and are essential tools for risk management and portfolio construction.

The sixth objective involved **investor cohort analysis** to understand how different groups of investors behave over time based on their entry characteristics. This analysis provides insights into investment patterns, retention rates, and the factors influencing investor decisions.

The seventh objective focused on **SIP continuity analysis** to identify investors at risk of discontinuing their systematic investment plans. Understanding discontinuation patterns enables proactive retention strategies and informs product development efforts.

The eighth objective required the development of a **mutual fund recommendation system** that matches investors with suitable schemes based on their risk appetite categories. This system translates complex analytical findings into practical, personalised investment guidance.

The ninth objective involved **portfolio concentration assessment using the Herfindahl-Hirschman Index (HHI)** to measure sector-level diversification within fund portfolios. This analysis helps investors understand the degree of concentration risk in their holdings.

The final objective encompassed the **development of interactive Power BI dashboards** that consolidate key findings and enable dynamic exploration of the mutual fund data. These dashboards serve as the primary interface for stakeholders to access and interact with the analytical outputs of the project.

3. Data Sources

The analytical framework developed in this project integrates data from nine distinct datasets, each providing a unique perspective on the mutual fund ecosystem. The diversity of these data sources enables multidimensional analysis, combining fund-level characteristics, historical performance, investor behaviour, industry trends, and portfolio composition. This section describes each dataset, its purpose within the analytical framework, and the key variables it contributes to the overall analysis.

3.1 Fund Master Data

The Fund Master Data serves as the foundational reference dataset containing comprehensive information about each mutual fund scheme. This dataset includes essential attributes such as the unique scheme code, scheme name, Asset Management Company (AMC) name, fund category, fund type, risk grade as defined by the rating agency, expense ratio, launch date, and current assets under management. The Fund Master Data functions as the primary dimension table, providing the descriptive context necessary for interpreting all other quantitative datasets. Without this master reference, numerical data from other sources would lack the categorical structure required for meaningful segmentation and comparison.

3.2 NAV History Data

The Net Asset Value (NAV) History Data contains daily NAV recordings for each scheme over an extended historical period. NAV represents the per-unit market value of a mutual fund scheme and is computed by dividing the total value of all securities in the portfolio, less liabilities, by the number of outstanding units. This dataset is fundamental for computing returns, volatility measures, drawdowns, and all performance metrics that depend on price movements over time. The granularity of daily NAV data enables both short-term and long-term trend analysis, supporting computations ranging from daily return calculations to annualised performance assessments.

3.3 Investor Transactions Data

The Investor Transactions Data captures detailed records of individual investor activities, including purchases, redemptions, and systematic investment plan (SIP) registrations. This dataset includes variables such as investor identifier, transaction date, transaction type, amount, scheme code, and investor demographic information where available. The transaction data is essential for analysing investor behaviour patterns, computing retention rates, identifying at-risk investors, and understanding the flow of money into and out of different scheme categories. It also supports the cohort analysis and SIP continuity analysis objectives of the project.

3.4 Scheme Performance Data

The Scheme Performance Data aggregates key performance metrics for each mutual fund scheme, including returns over various time horizons (one month, three months, six months, one year, three years, five years), standard deviation, Sharpe Ratio, Alpha, Beta, and Sortino Ratio. This dataset provides the quantitative foundation for comparing funds within and across categories, identifying top performers, and

evaluating risk-adjusted returns. The performance metrics contained in this dataset represent standardised measures that enable objective, apples-to-apples comparisons between schemes with different investment objectives and risk profiles.

3.5 Monthly SIP Inflows Data

The Monthly SIP Inflows Data tracks the aggregate systematic investment plan contributions received by each scheme on a monthly basis. This dataset reveals patterns in retail investor participation, as SIPs represent the preferred investment mode for individual investors seeking to build wealth systematically through rupee cost averaging. Analysing SIP inflow trends provides insights into investor confidence, the popularity of different fund categories, and the stability of funding sources for fund managers. Declining SIP inflows may signal deteriorating investor sentiment, while sustained growth indicates continued faith in long-term wealth creation through mutual funds.

3.6 Category Inflows Data

The Category Inflows Data aggregates investment flows at the fund category level, showing how capital moves between different types of mutual funds, including equity large-cap, equity mid-cap, equity smallcap, debt funds, hybrid funds, and others. This dataset enables analysis of sector rotation patterns, risk appetite shifts among investors, and the relative attractiveness of different asset classes over time. Category-level flow analysis is particularly valuable for understanding macro-level investment trends and identifying which segments of the mutual fund industry are experiencing growth or contraction.

3.7 Portfolio Holdings Data

The Portfolio Holdings Data provides detailed information about the securities held within each mutual fund scheme's portfolio. This includes stock names, sector classifications, percentage allocation, and market value of individual holdings. This dataset is essential for portfolio analysis, sector allocation assessment, and concentration risk measurement using metrics such as the Herfindahl-Hirschman Index. Understanding portfolio composition enables investors to evaluate whether a fund's actual holdings align with its stated investment objective and to assess the degree of diversification within the portfolio.

3.8 Benchmark Indices Data

The Benchmark Indices Data contains historical values for the market indices against which mutual fund schemes measure their performance. Common benchmarks include the Nifty 50, Nifty Midcap 100, Nifty Smallcap 250, and various debt indices. This dataset provides the market reference necessary for computing relative performance metrics such as Alpha and Beta, which indicate whether a fund has outperformed or underperformed the market after adjusting for risk. Benchmark data transforms absolute return figures into meaningful performance assessments by providing context about broader market movements during the same period.

3.9 Industry Folio Count Data

The Industry Folio Count Data tracks the number of investor folios (accounts) across the mutual fund industry over time, broken down by AMC, category, and investor type. Folio count serves as a proxy for retail participation and investor base expansion. This dataset supports trend analysis of industry growth, market share analysis among competing AMCs, and the assessment of financial inclusion efforts. Changes in folio counts reflect the success of distribution strategies, investor education initiatives, and the overall attractiveness of mutual funds as an investment vehicle for the mass market.

4. ETL Design and Data Preparation

The Extract-Transform-Load (ETL) process forms the backbone of any data analytics project, serving as the critical infrastructure that converts raw, heterogeneous data into a clean, consistent, and analysis-ready format. The ETL pipeline developed for this project was designed to handle the nine distinct mutual fund datasets described in the preceding section, each with unique structural characteristics, data quality profiles, and update frequencies. The pipeline was implemented using Python and the Pandas library within the Jupyter Notebook environment, ensuring reproducibility, documentation, and ease of maintenance.

The ETL architecture follows a three-phase sequential processing model comprising the Extract Phase, Transform Phase, and Load Phase. Each phase encompasses specific operations designed to progressively improve data quality and prepare the information for downstream analytical consumption. The workflow is illustrated in Figure 1, which provides a visual representation of the data flow from source files through transformation operations to final output datasets.

4.1 Extract Phase

The Extract Phase involves reading raw data files from their source locations and loading them into memory as structured dataframes. For this project, all source data was provided in Comma-Separated Values (CSV) format, which is a widely supported, platform-independent file format suitable for tabular data exchange. The extraction process was implemented using the Pandas `read_csv` function, which offers extensive configuration options for handling different encoding schemes, delimiter characters, date parsing, and missing value representations.

During extraction, each CSV file was loaded with appropriate parameters to ensure correct interpretation of the data. Date columns were parsed into datetime objects to enable temporal analysis, categorical columns were designated with appropriate data types to optimise memory usage, and column names were standardised to follow consistent naming conventions. The extraction process also generated summary statistics including row counts, column counts, and data type distributions for each dataset, providing an immediate overview of the data volume and structure.

Error handling mechanisms were implemented to manage potential issues during extraction, such as missing files, encoding mismatches, or malformed records. In cases where data quality issues were detected at the extraction stage, the issues were logged for resolution before proceeding to the transformation phase. This proactive approach prevented the propagation of data quality problems into downstream analytical processes.

4.2 Transform Phase

The Transform Phase represents the most computationally intensive and analytically critical component of the ETL pipeline. This phase encompasses all data cleaning, standardisation, enrichment, and restructuring operations necessary to convert raw extracted data into analysis-ready datasets. The transformation operations were organised into logical categories addressing specific data quality dimensions.

Handling Missing Values was the first transformation priority. Missing values were identified through systematic profiling of each dataset, quantifying the extent of missingness by column and by row. The treatment strategy varied depending on the nature of the missing data and the analytical importance of the affected variable. For numerical columns with a small percentage of missing values, median imputation was applied to preserve the distributional characteristics of the data. For categorical columns, missing values were either filled with a designated "Unknown" category or with the mode value depending on the context. In cases where the proportion of missing values exceeded a defined threshold, the column was flagged for potential exclusion from certain analyses. Rows with critical missing values in key identifier fields were quarantined for manual review.

Removing Duplicates was essential to prevent analytical bias and ensure accurate aggregation. Duplicate records were identified based on business key combinations such as scheme code plus date for NAV data, or investor identifier plus transaction date for transaction data. The deduplication process retained the first occurrence of each unique record and removed subsequent duplicates, with the removal counts logged for audit purposes. Special attention was given to near-duplicate records that might differ only in timestamp precision or formatting, which were standardised before deduplication.

Standardising Column Names ensured consistency across all datasets and facilitated subsequent integration operations. A unified naming convention was applied, converting all column names to lowercase, replacing spaces with underscores, and removing special characters. For example, "Scheme Code" became "scheme_code", "NAV Date" became "nav_date", and "AUM (Cr)" became "aum_cr". This standardisation eliminated the need for column name mapping in downstream code and improved the readability and maintainability of the analytical scripts.

Converting Data Types optimised memory usage and ensured correct computational behaviour. Numeric columns stored as strings were converted to appropriate numeric types (integer or float). Date strings were parsed into datetime objects. Categorical columns with a limited set of possible values were converted to Pandas categorical data types, reducing memory footprint and improving processing speed. Currency values were standardised to consistent units (typically crores of Indian Rupees) to prevent unit mismatch errors in calculations.

Feature Engineering created new derived variables that enhance analytical capability. Examples include computing scheme age from the launch date, categorising funds into risk buckets based on their risk grade, calculating AUM quartiles for relative size classification, and creating time-based features such as month, quarter, and year from date fields. These engineered features serve as dimensions for segmentation and filtering in both statistical analysis and dashboard visualisation.

Dataset Integration established the relationships between the various data sources through key-based joins. The Fund Master Data served as the central dimension table, linked to NAV History via scheme code, to Performance Data via scheme code, to Transaction Data via scheme code, and to Portfolio Holdings via scheme code. Benchmark Indices were linked based on benchmark name or code. These integrated views enable multidimensional analysis that combines descriptive fund characteristics with quantitative performance and behavioural data.

4.3 Load Phase

The Load Phase involves persisting the transformed datasets to storage in a structured format that supports efficient retrieval for analysis. Each processed dataset was saved as a CSV file with a clear naming convention indicating the dataset name and processing date. The load process also generated a data dictionary documenting all columns, their data types, descriptions, and any transformations applied during the ETL process.

Quality assurance checks were performed after loading to verify that the output datasets met predefined quality criteria. These checks included row count validation, null value verification, date range confirmation, and referential integrity testing across related datasets. Any discrepancies were investigated and resolved before proceeding to the analysis phase. The ETL pipeline was designed to be fully reproducible, with all transformation logic documented in version-controlled Jupyter Notebooks, enabling future re-execution as source data is updated.

5. Exploratory Data Analysis

Exploratory Data Analysis (EDA) constitutes a critical phase in the analytical workflow, serving as the bridge between data preparation and formal statistical modelling. The EDA process involves systematically examining the datasets through descriptive statistics and visualisation techniques to uncover patterns, identify anomalies, test assumptions, and generate hypotheses. This section presents the findings of the exploratory analysis organised into six thematic areas: Investor Demographics, Mutual Fund Industry Trends, Fund Performance Analysis, Risk and Expense Analysis, NAV and Portfolio Analysis, and SIP Behaviour Analysis. Each subsection presents relevant visualisations with detailed interpretations of their analytical significance.

5.1 Investor Demographics

Understanding the demographic composition of the mutual fund investor base is essential for product development, marketing strategy, and investor education initiatives. The demographic analysis examined three key dimensions: age group distribution, gender distribution, and geographic distribution by city tier.

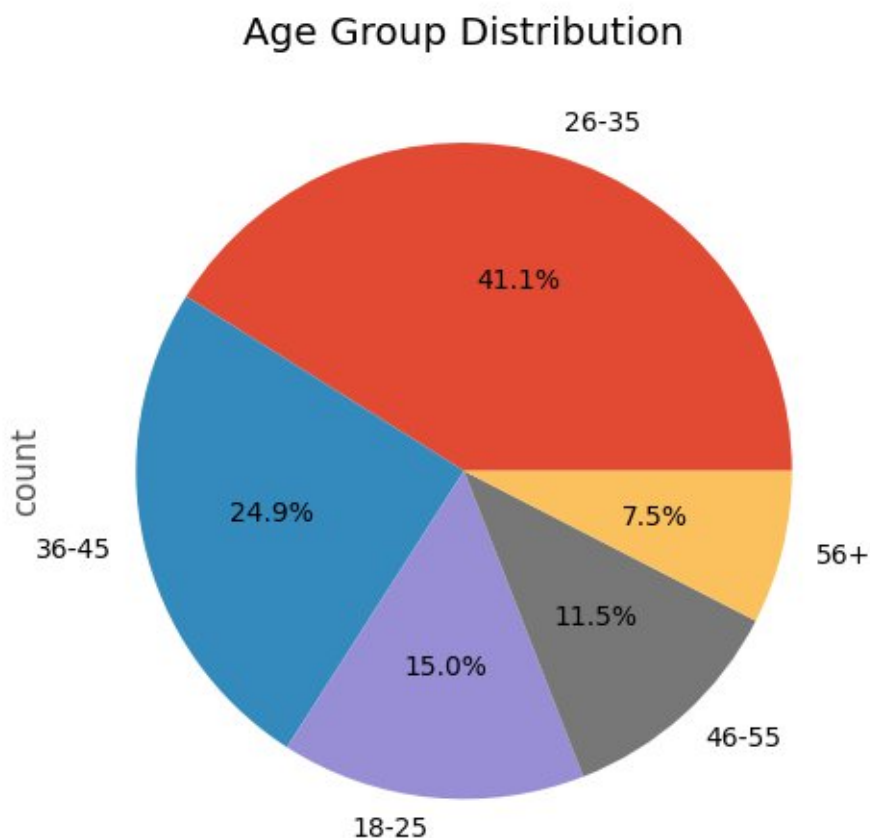


Figure 1:
Age Group Distribution of Mutual Fund Investors (age_group_distribution.png)

Figure 1 presents the age group distribution of mutual fund investors, revealing the generational composition of the participant base. The chart illustrates the relative proportion of investors across different age brackets, typically segmented into groups such as below 25, 25-35, 35-45, 45-55, 55-65, and above 65 years. Analysis of this distribution provides insights into the life-stage positioning of mutual

fund investors. A concentration in the 25-45 age range would indicate strong participation from young

professionals in the wealth accumulation phase, suggesting that mutual funds are successfully penetrating the millennial and Generation Z demographics. Conversely, a skew towards older age groups might indicate that mutual fund investing remains predominantly appealing to investors closer to retirement who possess greater investible surplus. The age distribution has significant implications for product design, as different age cohorts exhibit varying risk tolerances, investment horizons, and product preferences. Younger investors typically favour equity-oriented schemes with higher growth potential, while older investors may prefer debt or hybrid funds prioritising capital preservation and regular income.

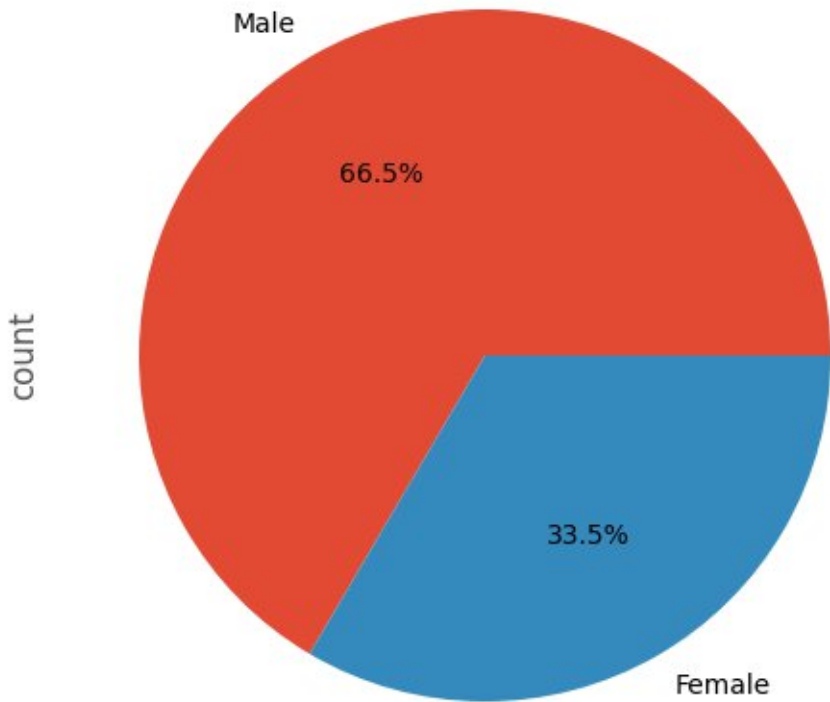


Figure 2: Gender Distribution of Mutual Fund Investors (gender_split.png)

Figure 2 displays the gender split among mutual fund investors, highlighting the degree of gender parity in retail mutual fund participation. The gender distribution is an important indicator of financial inclusion and the effectiveness of outreach efforts targeting underrepresented demographics. A significant gender imbalance would suggest the need for targeted investor education programmes, product positioning adjustments, and distribution channel strategies to engage the underrepresented gender more effectively. Research in behavioural finance has identified differences in risk perception, investment confidence, and decision-making patterns between male and female investors, making gender-aware product development and communication strategies increasingly relevant for Asset Management Companies seeking to broaden their investor base.

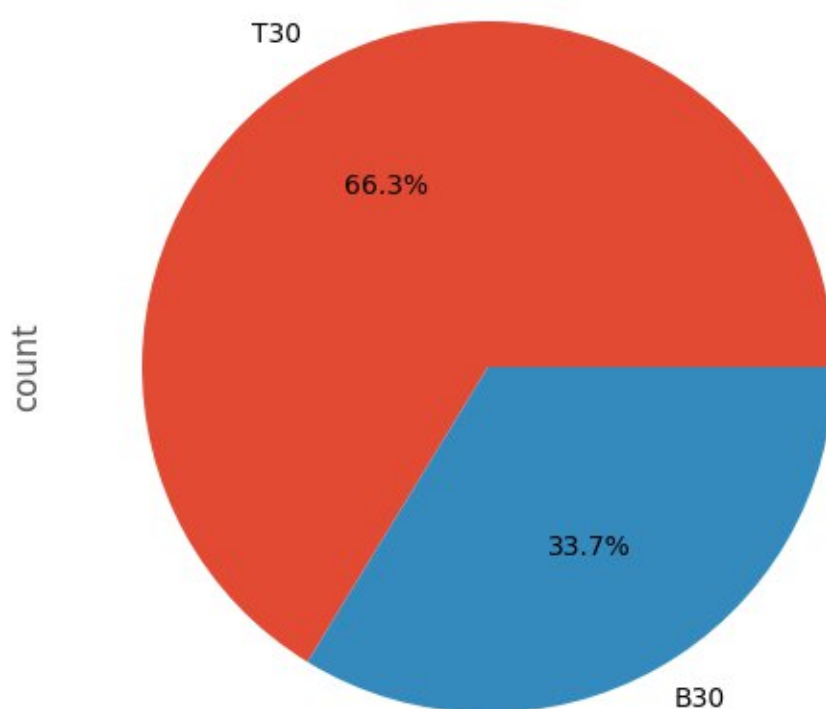


Figure 3: City Tier Distribution of Mutual Fund Investors (city_tier_distribution.png)

Figure 3 illustrates the geographic distribution of investors across city tiers, categorising locations into Tier 1 (major metropolitan areas), Tier 2 (developing urban centres), and Tier 3 (smaller cities and towns). This distribution reveals the penetration depth of mutual fund distribution networks beyond India's major financial centres. A heavy concentration in Tier 1 cities would indicate that mutual fund investing remains an urban, affluent phenomenon, while meaningful representation in Tier 2 and Tier 3 locations would demonstrate successful expansion into emerging markets. The city tier distribution has direct implications for distribution strategy, as Tier 2 and Tier 3 cities often require different engagement models including digital-first approaches, regional language support, and partnerships with local financial advisors.

5.2 Mutual Fund Industry Trends

The mutual fund industry trend analysis examines macro-level indicators that reflect the overall health and trajectory of the sector. Three key metrics were analysed: folio growth, SIP inflow trends, and category-wise inflow patterns.

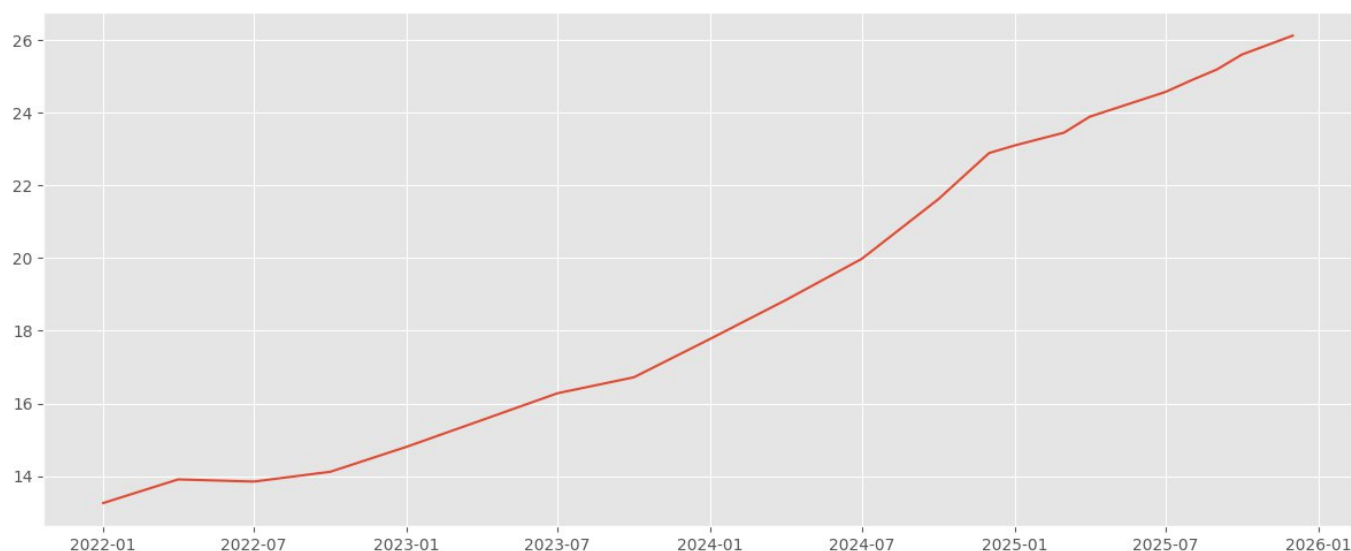


Figure 4: Mutual Fund Folio Growth Trend Over Time (folio_growth.png)

Figure 4 presents the trend in mutual fund folio counts over time, serving as a proxy for retail investor base expansion. The folio growth trajectory indicates the industry's success in attracting new investors and the stickiness of existing investor relationships. Sustained upward movement in folio counts suggests growing retail participation and improving financial penetration, while plateaus or declines may signal market saturation, competitive pressure from alternative investment products, or deteriorating investor sentiment. The rate of folio growth relative to AUM growth provides additional insight into whether

Industry expansion is driven by new investor acquisition or increased commitments from existing investors.

Monthly SIP Inflows

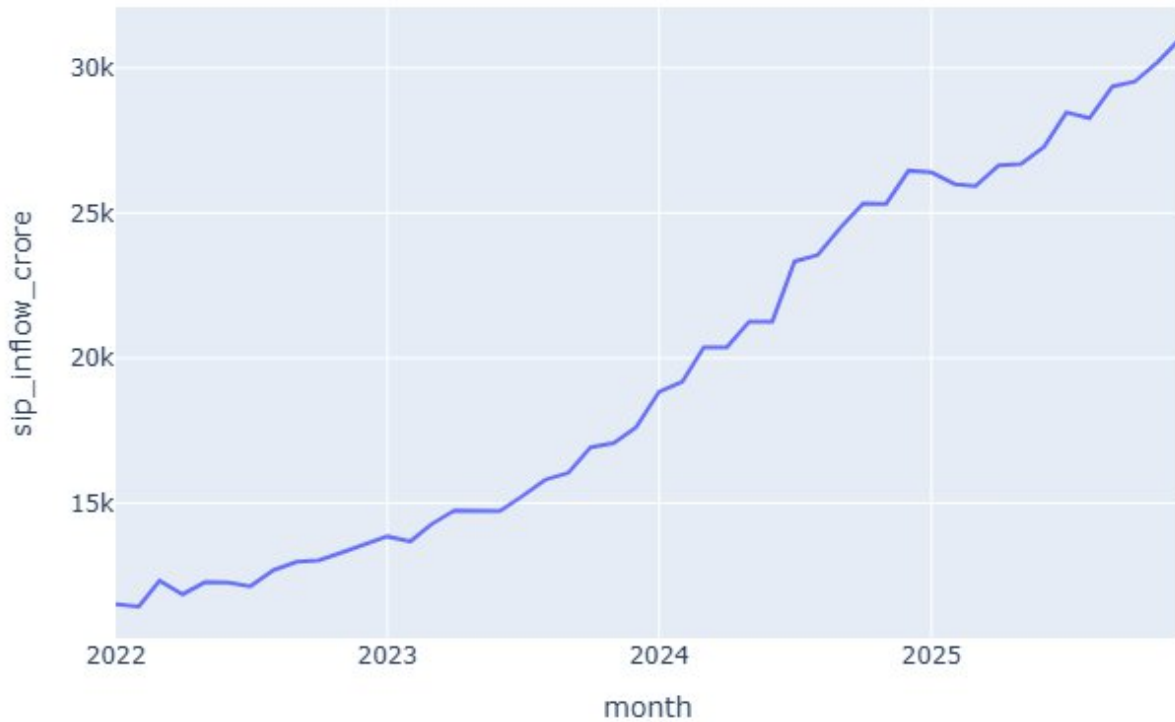


Figure 5: Monthly SIP Inflow Trend (sip_inflow_trend.png)

Figure 5 displays the monthly trend in SIP inflows, one of the most closely watched indicators in the Indian mutual fund industry. SIP inflows represent disciplined, recurring retail investments and are considered a stable, sticky source of funding for fund managers. The trend line reveals patterns in retail investor confidence and commitment to long-term wealth creation. Rising SIP inflows indicate growing acceptance of systematic investing as a wealth-building strategy and reflect positive investor sentiment towards equity markets. Seasonal patterns, if present, may align with tax-saving investment periods or festive seasons. Sharp declines in SIP inflows could signal panic during market corrections or economic uncertainty, while resilience during market downturns demonstrates investor maturity and commitment to long-term investment discipline.

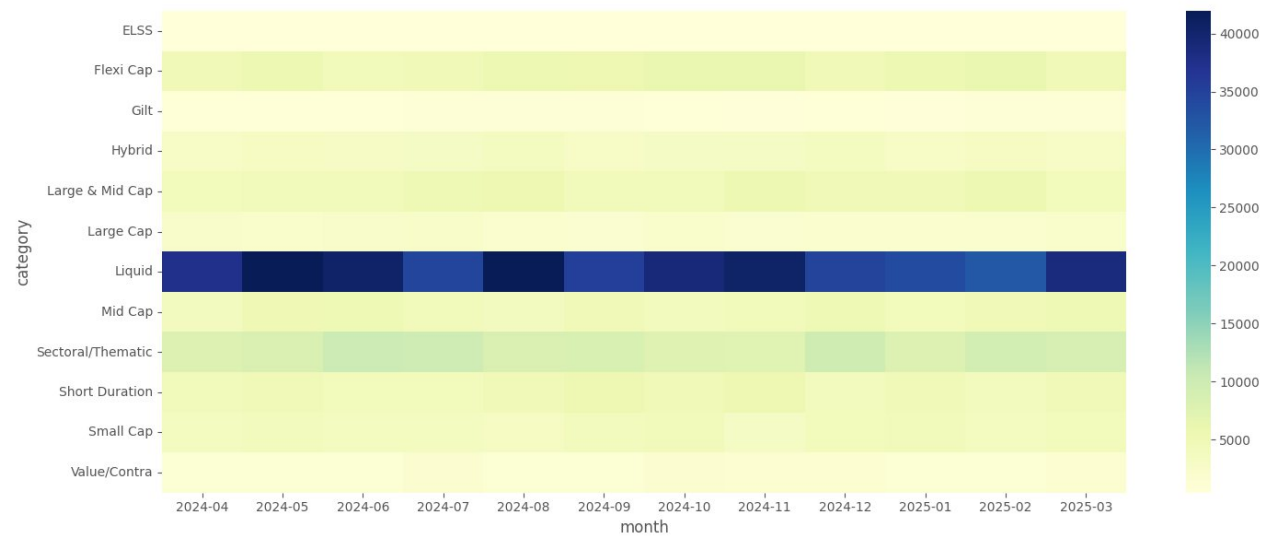


Figure 6: Category-wise Inflow Heatmap (category_inflow_heatmap.png)

Figure 6 presents a heatmap visualisation of investment inflows across mutual fund categories over time. This visualisation uses colour intensity to represent the magnitude of inflows, enabling rapid identification of hot and cold categories. The heatmap reveals investor preference shifts between asset classes, with bright areas indicating categories receiving strong inflows and darker areas showing categories experiencing outflows or weak inflows. Patterns in the heatmap may reveal sector rotation behaviour, risk appetite changes, and the impact of regulatory changes or market events on category preferences. For instance, a shift from small-cap to large-cap categories might indicate risk aversion during periods of market volatility, while strong flows into debt categories could signal a flight to safety during economic uncertainty.

5.3 Fund Performance Analysis

Performance analysis examines how mutual fund schemes have delivered returns relative to their peers and benchmarks. This analysis is fundamental to investment decision-making, as historical performance, while not guaranteeing future results, provides insight into fund management quality and strategy effectiveness.

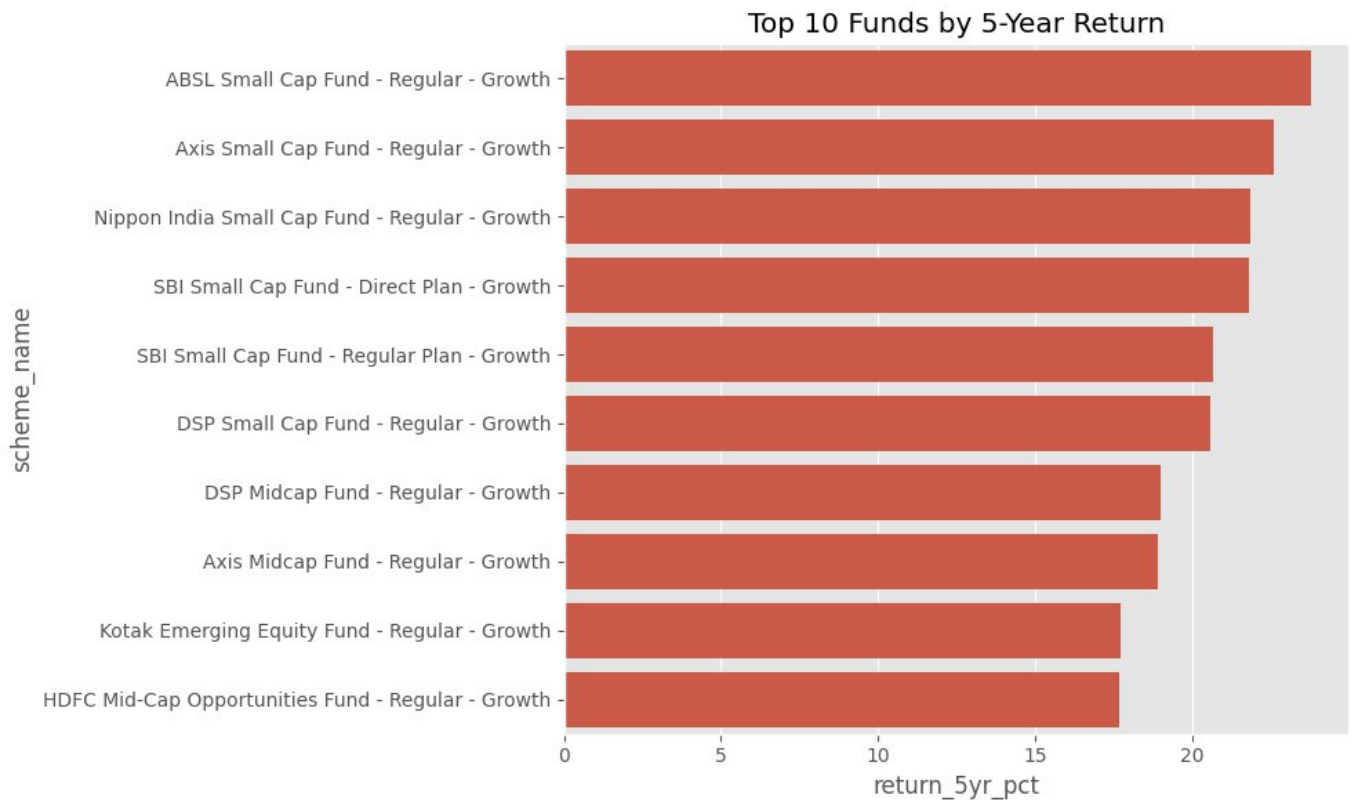


Figure 7: Top 10 Mutual Fund Schemes by Absolute Returns (top_10_returns.png)

Figure 7 ranks the top ten performing mutual fund schemes by their absolute returns over a specified time period. This chart identifies the schemes that have generated the highest nominal returns, attracting investor attention and competitive pressure. While absolute return rankings are popular among retail investors, they must be interpreted with caution as they do not account for the risk undertaken to achieve those returns. A fund generating exceptionally high returns may be concentrating in speculative segments, employing aggressive strategies, or benefiting from favourable market conditions that may not persist. The analysis of top performers should always be complemented with risk-adjusted metrics and an understanding of the investment philosophy and process that produced the returns.

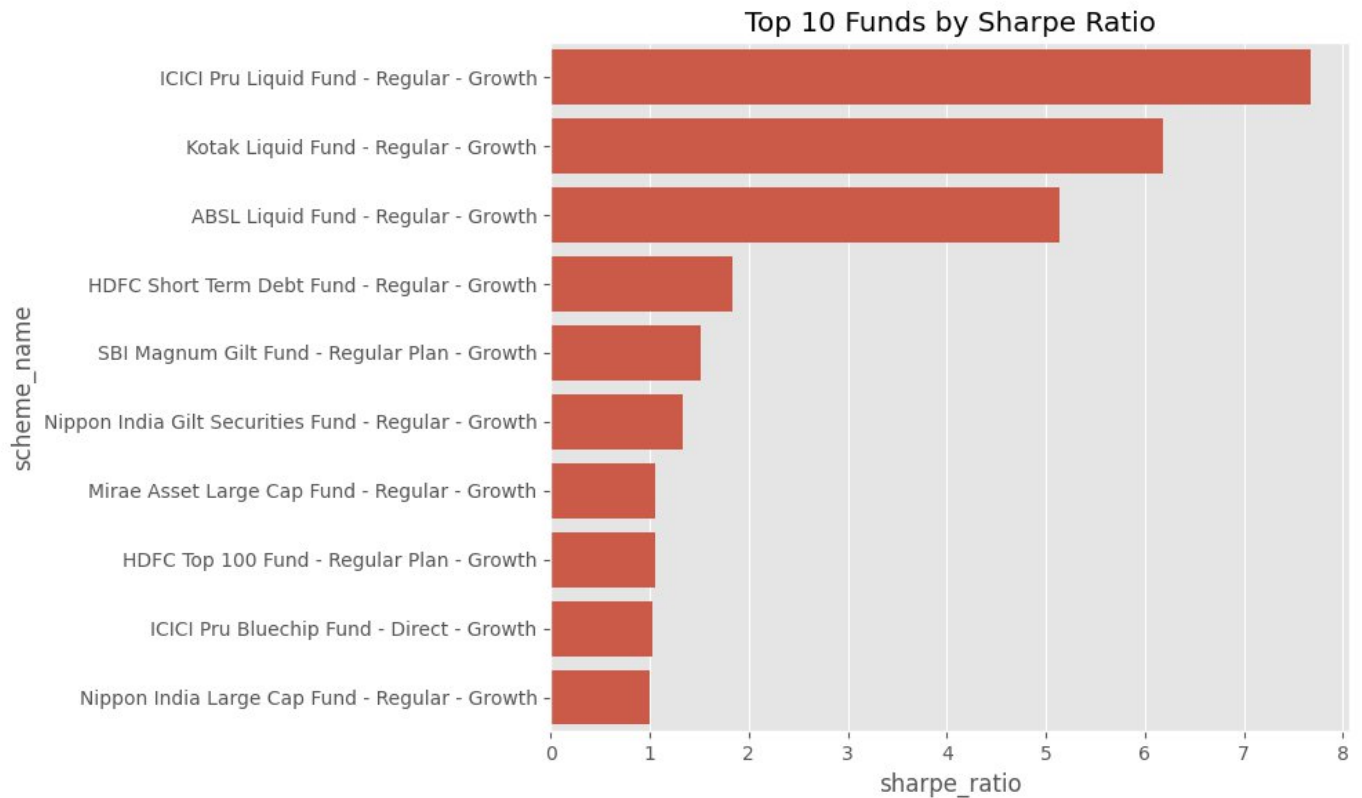


Figure 8: Top Performing Funds by Sharpe Ratio (top_sharpe_ratio.png)

Figure 8 presents the top-performing funds ranked by their Sharpe Ratio, which measures risk-adjusted returns by calculating the excess return per unit of total risk (standard deviation). Funds ranking highly on this metric have delivered superior returns relative to the volatility experienced by investors. The Sharpe Ratio provides a more nuanced performance assessment than absolute returns alone, penalising funds that achieve high returns through excessive risk-taking. This metric is particularly valuable for comparing funds across different categories and investment styles, as it standardises performance by risk. Funds appearing in the top Sharpe Ratio rankings but not in the top absolute return rankings typically represent disciplined, consistent performers that may be overlooked by investors focused solely on headline returns.

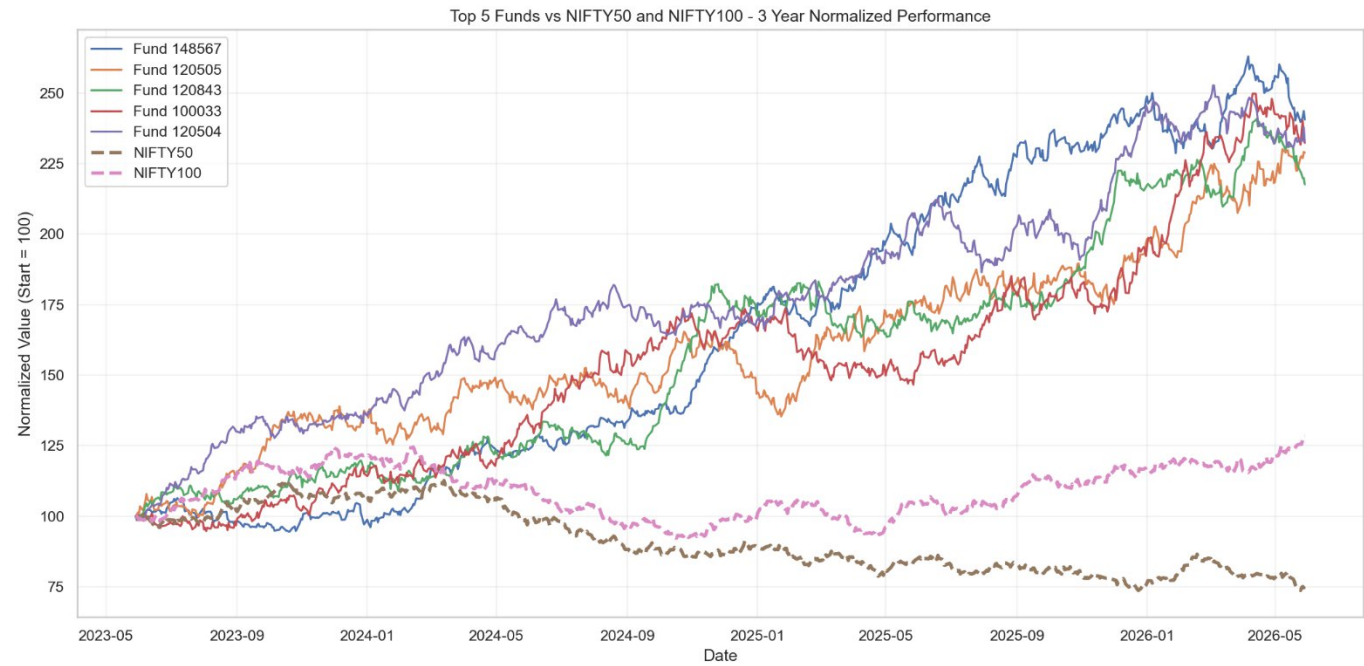


Figure 9: Fund Performance versus Benchmark Comparison (benchmark_comparison.png)

Figure 9 illustrates how mutual fund schemes have performed relative to their designated benchmarks. This comparison is fundamental to active management evaluation, as the primary justification for active management fees is the ability to consistently outperform the benchmark after accounting for costs. The visualisation typically shows the percentage of schemes outperforming their benchmarks within each category, along with the average margin of outperformance or underperformance. Categories with high outperformance rates suggest favourable conditions for active management, while categories with

widespread underperformance may indicate market efficiency that makes consistent outperformance difficult. This analysis helps investors set realistic expectations and make informed decisions about active versus passive investment approaches.

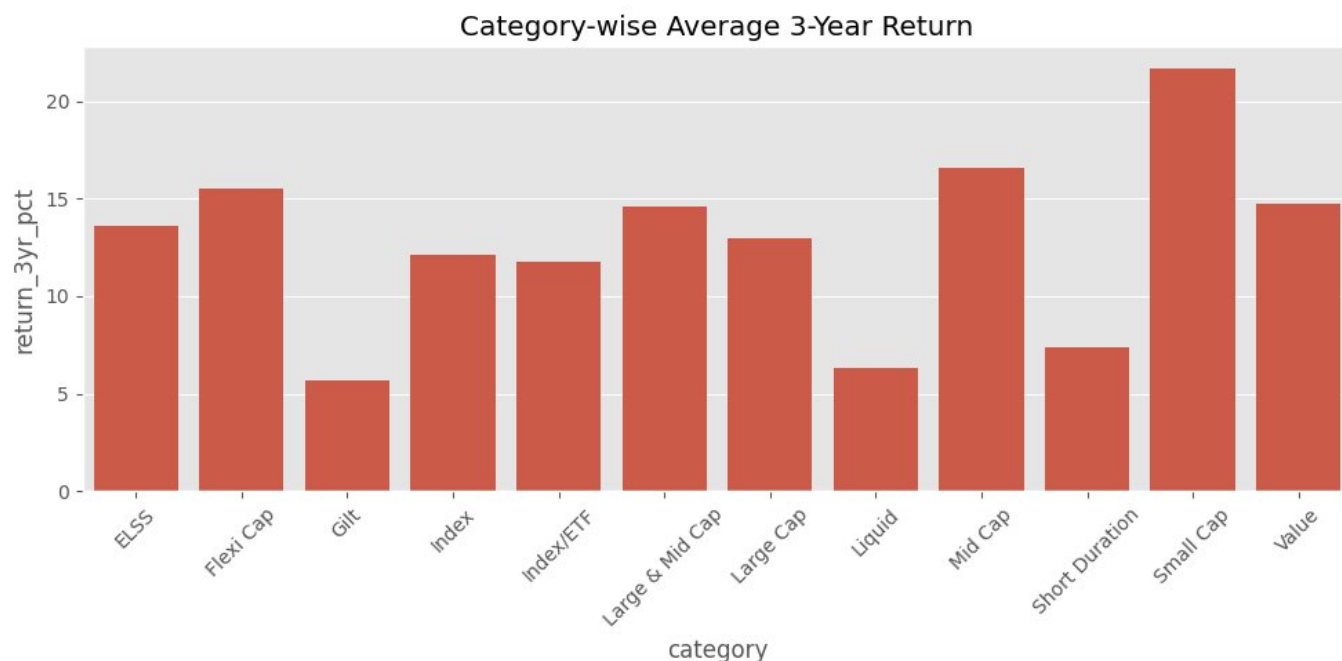


Figure 10: Average Returns by Fund Category (category_avg_return.png)

Figure 10 displays the average returns achieved by each mutual fund category over a specified time horizon. This category-level view enables investors to understand which segments of the market have been the most rewarding and provides context for evaluating individual fund performance. A fund outperforming the category average may indicate superior management, while consistent underperformance relative to the category average raises questions about the fund's investment approach or execution. Category average returns also reveal the risk-return trade-offs inherent in different asset classes, with equity categories typically showing higher average returns accompanied by greater volatility, and debt categories offering more modest but stable returns.

5.4 Risk and Expense Analysis

Risk and expense analysis examines the risk profiles and cost structures of mutual fund schemes. Understanding these dimensions is essential for making informed investment decisions, as risk determines the potential for loss and expenses directly impact net returns.

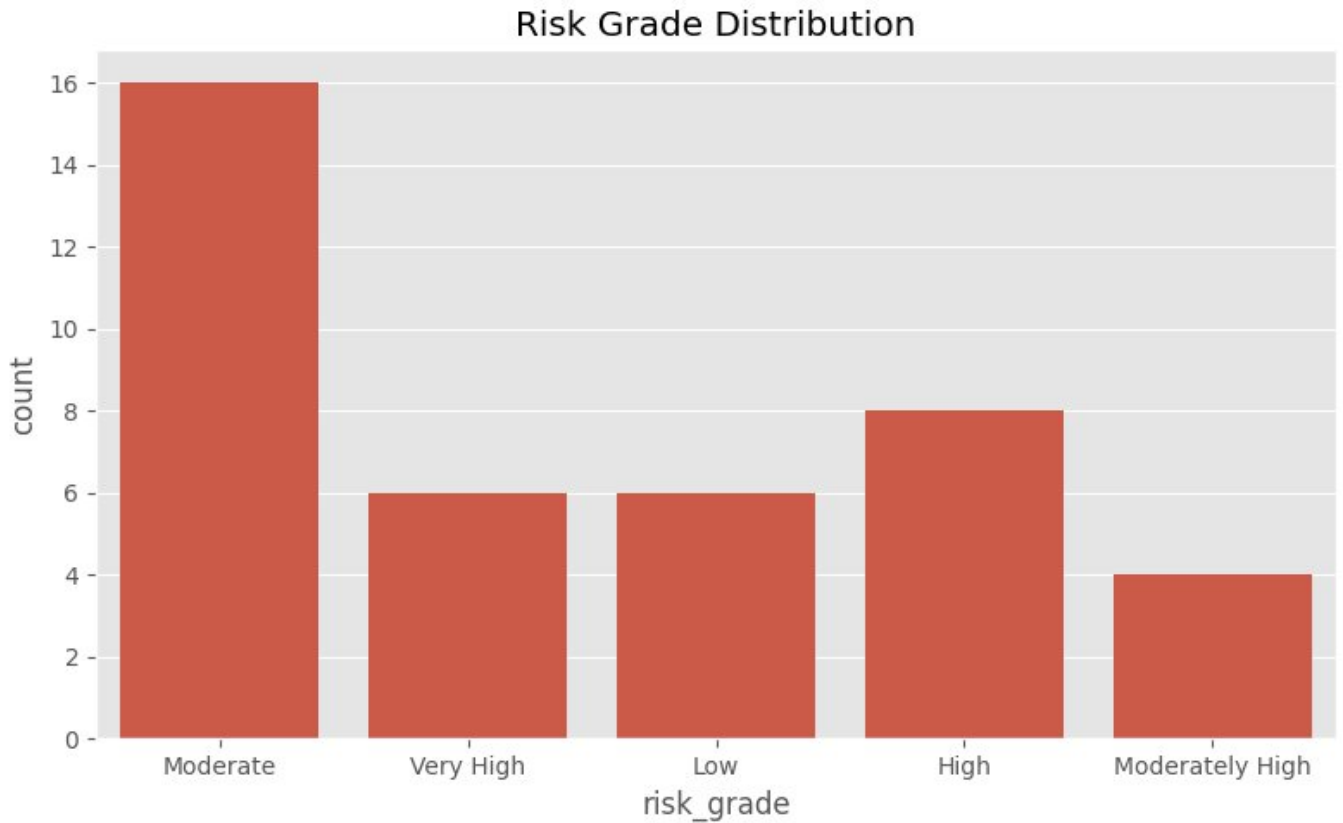


Figure 11: Distribution of Funds Across Risk Grades (risk_grade_distribution.png)

Figure 11 shows the distribution of mutual fund schemes across different risk grades assigned by rating agencies. Risk grades typically range from low to very high, reflecting the potential for investment value fluctuation and capital loss. The distribution reveals the risk profile of the overall mutual fund universe and the availability of options for investors with different risk tolerances. A balanced distribution across risk grades indicates a healthy ecosystem catering to diverse investor needs, while concentration in high-risk categories might suggest a market skewed towards aggressive investors. For individual investors, understanding the risk grade distribution helps in constructing portfolios that align with their personal risk capacity and financial goals.

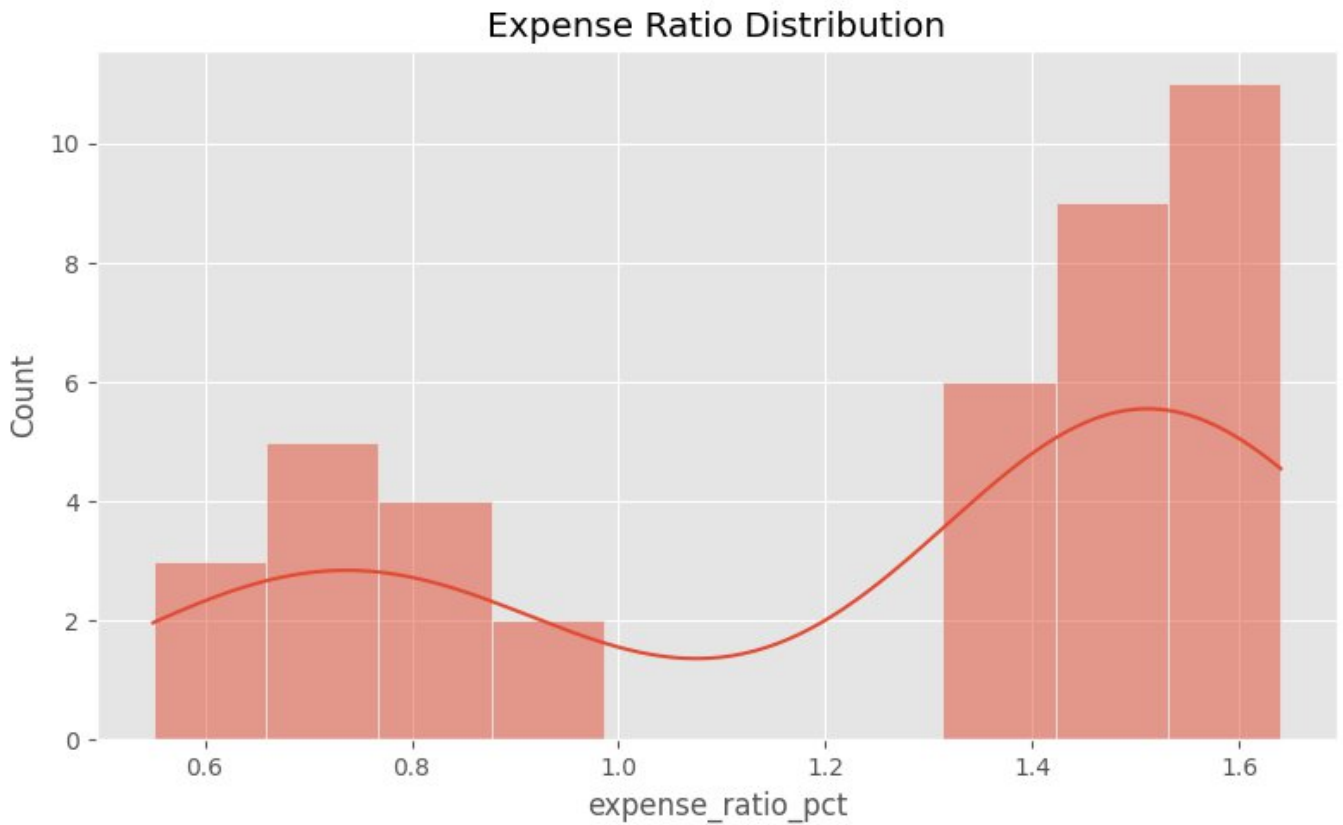


Figure 12: Expense Ratio Distribution Across Mutual Fund Schemes (expense_ratio_distribution.png)

Figure 12 presents the distribution of expense ratios across mutual fund schemes. The expense ratio represents the annual cost of managing and operating the fund, expressed as a percentage of average assets under management. This fee directly reduces the net return received by investors, making expense ratio analysis a critical component of fund selection. The distribution reveals the range of costs within and across categories, highlighting which categories tend to have higher or lower expense structures. Equity funds typically carry higher expense ratios than debt funds due to greater research and transaction costs. Variations in expense ratios within the same category may reflect differences in fund size (larger funds often benefit from economies of scale), distribution arrangements, or the complexity of the investment strategy.

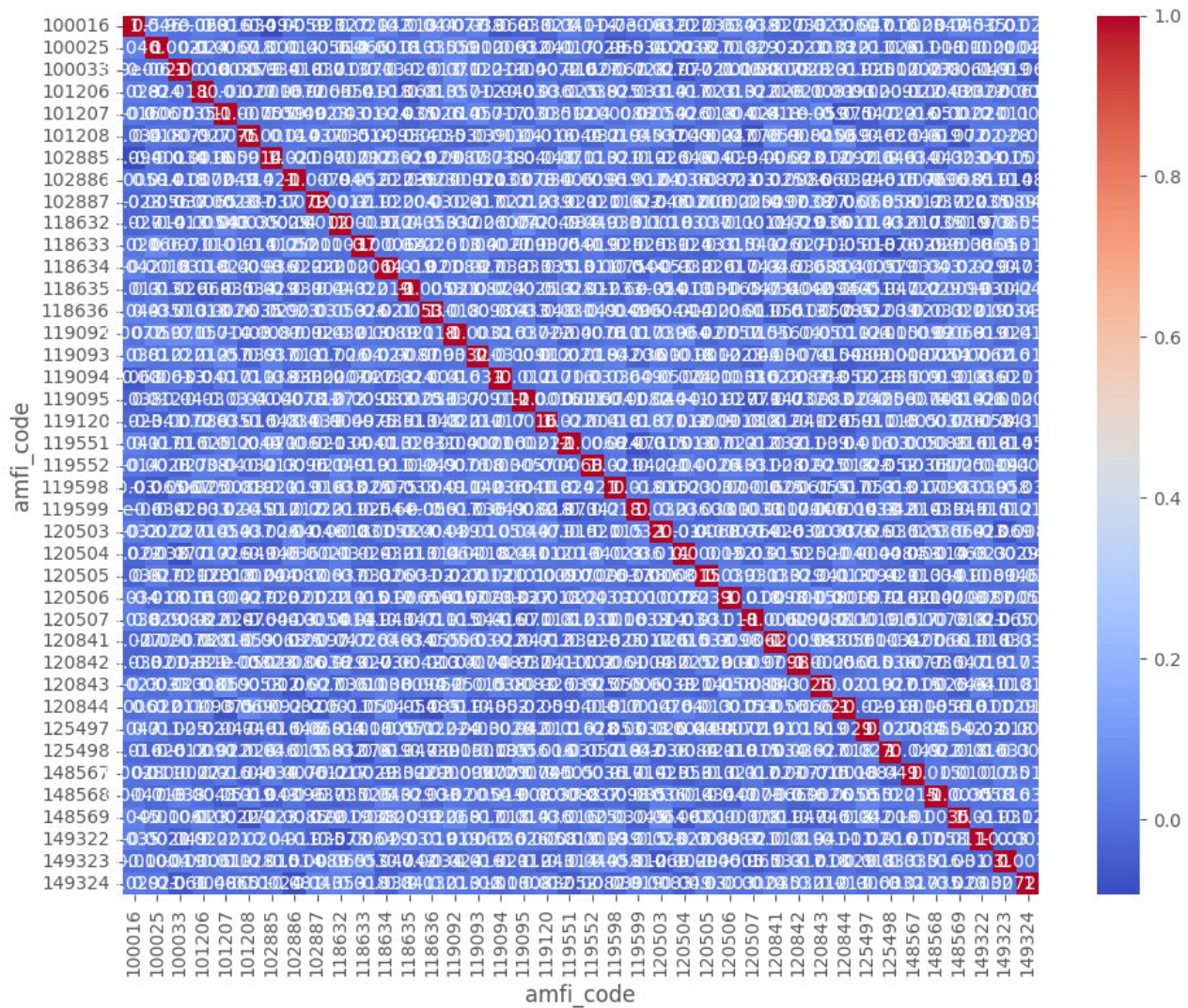


Figure 13: NAV Correlation Matrix Across Fund Categories (nav_correlation_matrix.png)

Figure 13 displays the correlation matrix of NAV movements across different mutual fund categories. Correlation measures the degree to which two variables move in relation to each other, with values ranging from negative one (perfect inverse movement) to positive one (perfect parallel movement). In portfolio construction, understanding correlations is essential for diversification, as combining assets with low or negative correlations can reduce overall portfolio volatility without sacrificing expected returns. The correlation matrix reveals which categories tend to move together and which offer diversification benefits. High correlations between categories suggest limited diversification value from combining them, while low correlations indicate potential risk reduction through strategic allocation across uncorrelated segments.

5.5 NAV and Portfolio Analysis

NAV trend analysis and portfolio composition examination provide insights into fund valuation trajectories and the underlying asset allocation decisions made by fund managers.

Daily NAV Trends (2022-2026)

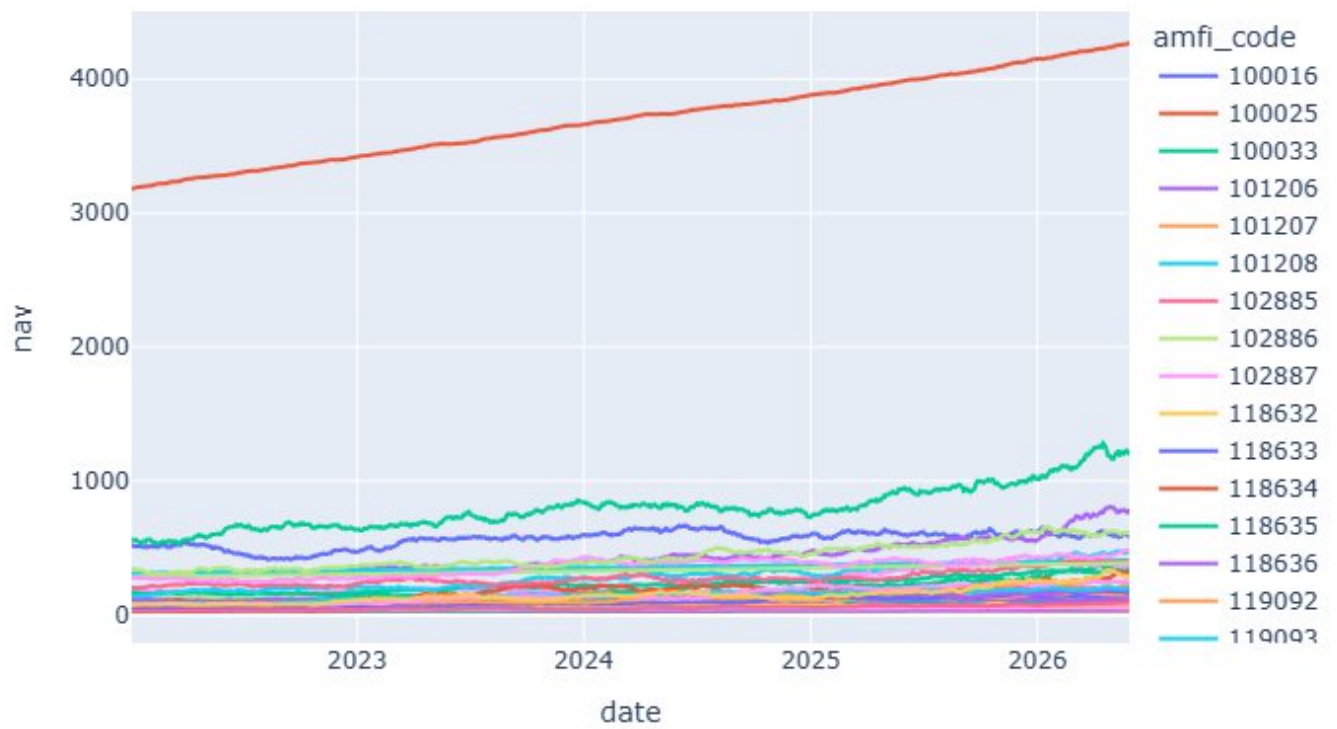


Figure 14: Net Asset Value Trend Over Time (nav_trend.png)

Figure 14 illustrates the NAV trend for selected mutual fund schemes over a historical period. The NAV trajectory reflects the cumulative impact of the fund's investment decisions, market movements, and income distributions. Analysing NAV trends enables investors to visualise the growth path of their investments and understand the volatility characteristics of different schemes. Steep, consistent upward trends indicate successful wealth creation, while highly volatile trends suggest greater uncertainty in returns. Comparing NAV trends across multiple schemes or against benchmark indices provides visual context for performance evaluation and helps investors align their expectations with historical patterns.

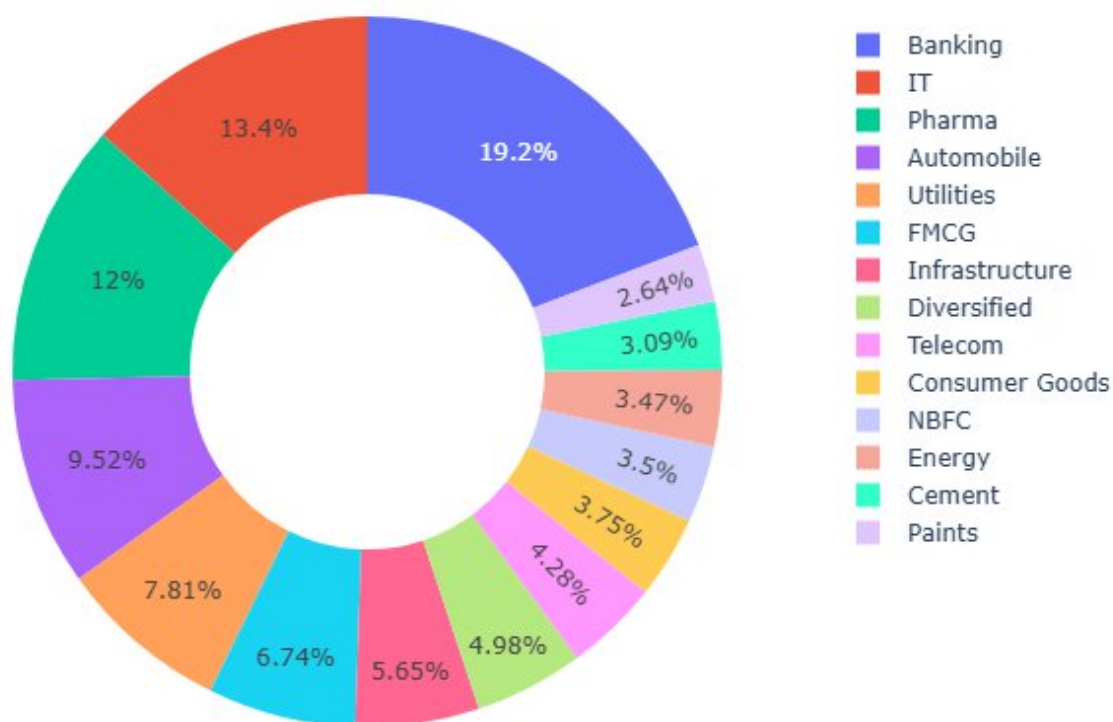


Figure 15: Sector Allocation of Mutual Fund Portfolio (sector_allocation.png)

Figure 15 presents the sector allocation of a mutual fund's portfolio, showing how investments are distributed across different industry sectors such as banking and financial services, information technology, healthcare, consumer goods, energy, and manufacturing. Sector allocation reveals the fund manager's views on sectoral growth prospects and risk assessment. A well-diversified portfolio spans multiple sectors, reducing concentration risk from overexposure to any single industry. Conversely, concentrated sector bets may generate outperformance if the favoured sectors perform well but increase vulnerability to sector-specific downturns. Comparing a fund's sector allocation to its benchmark reveals active sector overweights and underweights, providing insight into the manager's tactical positioning.

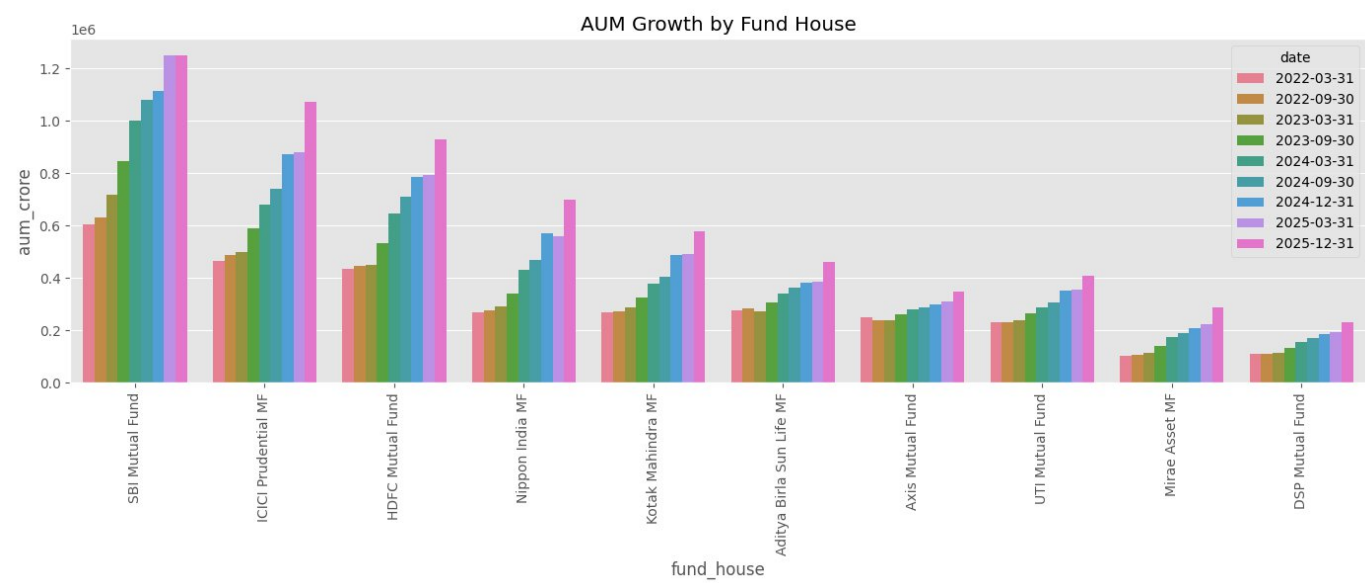


Figure 16: Assets Under Management Growth Trend (aum_growth.png)

Figure 16 displays the AUM growth trend for selected mutual fund schemes or the industry as a whole. AUM growth reflects the net effect of investment returns and investor flows (new subscriptions minus redemptions). Consistent AUM growth indicates positive investor sentiment and successful performance, while declining AUM may signal poor performance, unfavourable market conditions, or increased competition. AUM trends also have operational implications, as funds below certain size thresholds may face viability challenges due to inadequate revenue to cover fixed operating costs. For investors, sustained AUM growth in a well-performing fund confirms broader market confidence in the scheme's management.

5.6 SIP Behaviour Analysis

SIP behaviour analysis examines patterns in systematic investment plan contributions, which represent the primary investment vehicle for retail mutual fund participants in India.

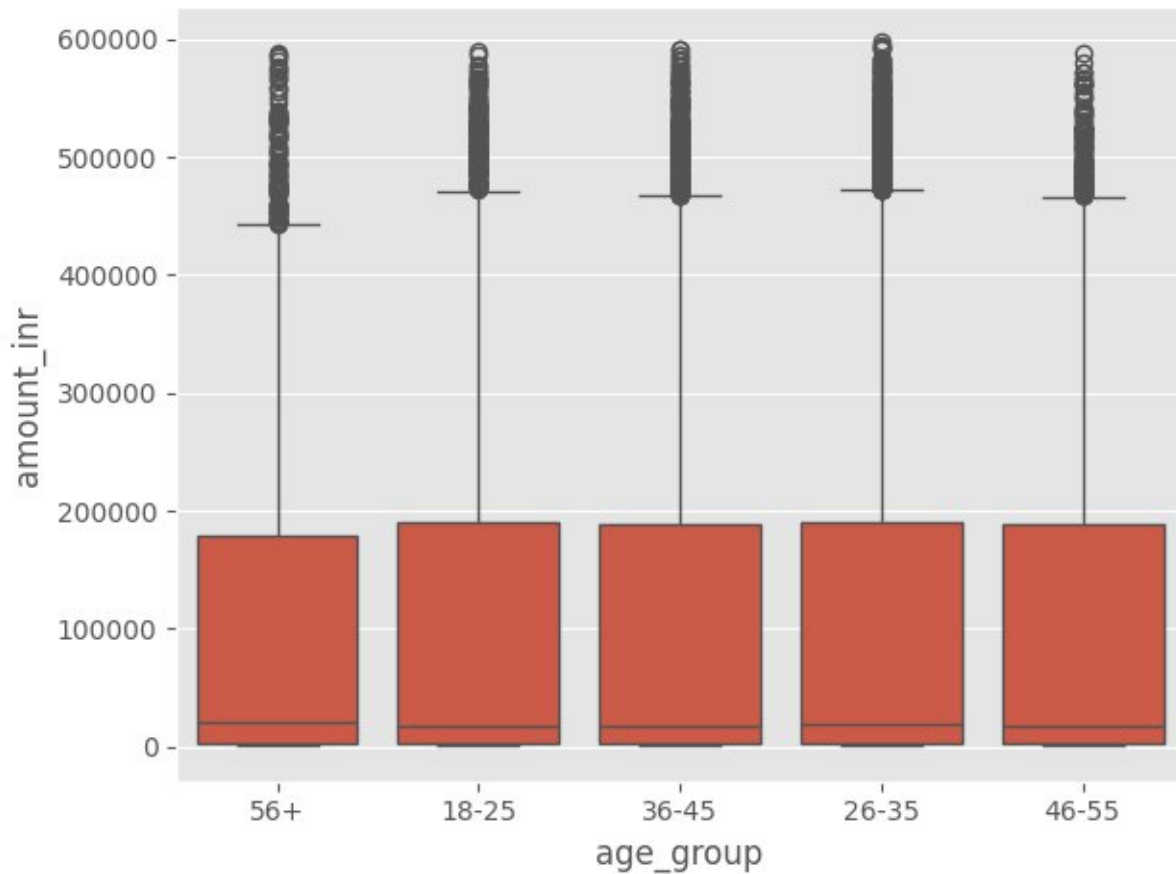


Figure 17: Distribution of SIP Investment Amounts (sip_amount_boxplot.png)

Figure 17 presents a boxplot visualisation of SIP investment amounts, displaying the distribution's median, quartiles, and outliers. The boxplot format effectively reveals the central tendency and spread of SIP contributions, as well as the presence of unusually high or low values. Analysis of the SIP amount distribution provides insights into the affordability positioning of mutual fund products and the typical commitment levels of retail investors. A median in the range of one to five thousand rupees suggests strong participation from middle-income investors, while a long tail of higher values indicates significant participation from affluent investors making larger systematic commitments. The interquartile range reveals the diversity of investment amounts, with a narrow range suggesting homogeneous investor profiles and a wide range indicating diverse participation across income segments.

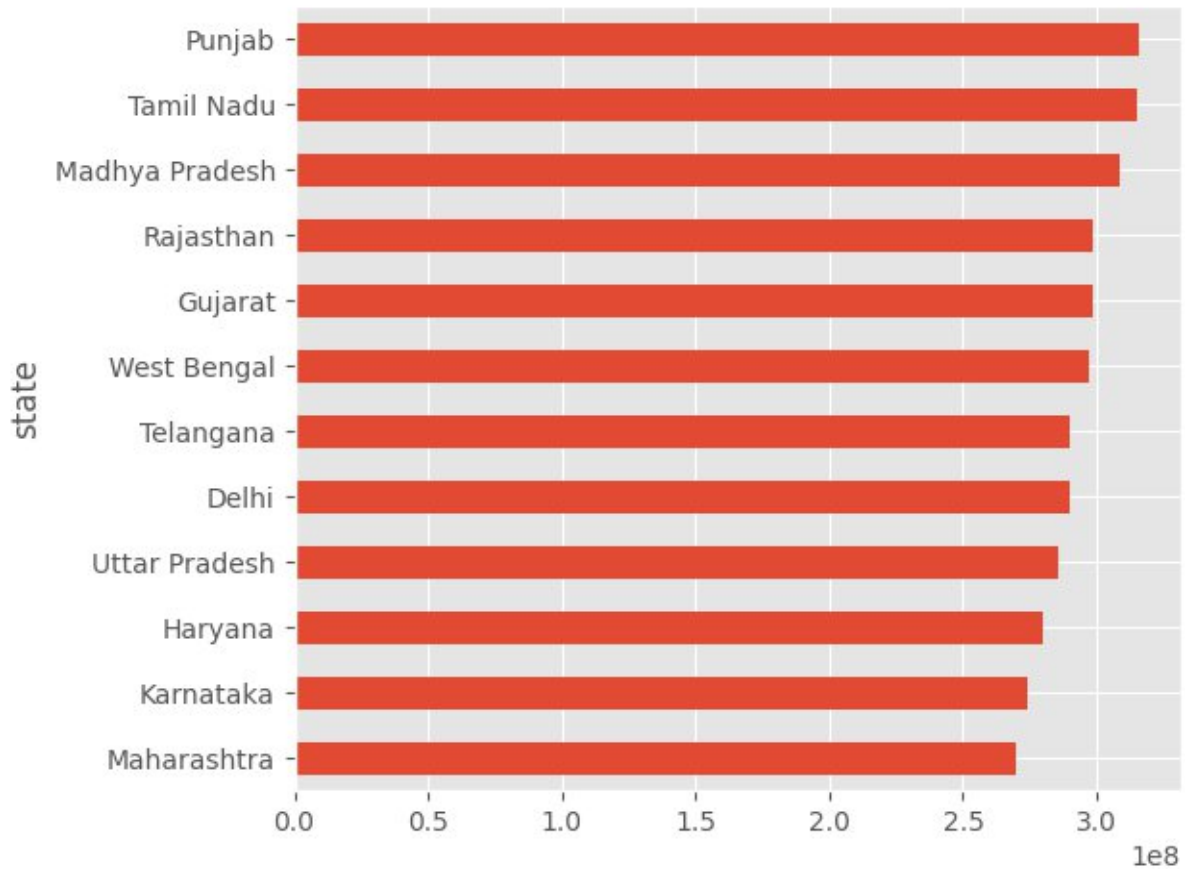


Figure 18: State-wise SIP Investment Amounts (state_sip_amount.png)

Figure 18 illustrates the geographic distribution of SIP investments across Indian states, revealing regional variations in mutual fund participation and average contribution levels. This geographic analysis identifies states with the highest SIP penetration and those with growth potential. States with high SIP amounts typically correlate with higher per capita income, greater financial literacy, and more developed distribution infrastructure. Conversely, states with lower SIP participation represent opportunities for market expansion through targeted investor education and distribution development. Understanding state-wise patterns enables Asset Management Companies to allocate marketing resources efficiently and tailor their outreach strategies to regional preferences and characteristics.

5.7 Summary of EDA Findings

The exploratory data analysis conducted across six thematic dimensions has yielded several important findings that inform the subsequent analytical phases and provide context for the overall project outcomes.

From the demographic analysis, the mutual fund investor base in India demonstrates a composition that reflects both the growing financial awareness among younger professionals and the sustained participation of established investors. The geographic distribution reveals significant potential for expansion beyond major metropolitan centres into Tier 2 and Tier 3 cities, where rising incomes and increasing digital penetration are creating favourable conditions for mutual fund adoption.

Industry trend analysis confirms the growing popularity of systematic investment plans as the preferred mode of retail participation, with SIP inflows demonstrating resilience and consistency even during periods of market volatility. This trend indicates a maturing investor base that understands the benefits of disciplined, long-term investing through rupee cost averaging. Category flow patterns reveal dynamic investor preferences that shift in response to market conditions, regulatory changes, and evolving investment objectives.

Performance analysis highlights the importance of using risk-adjusted metrics rather than absolute returns for fund evaluation. Funds generating the highest absolute returns do not necessarily deliver the best risk-adjusted outcomes, emphasising the need for comprehensive performance assessment frameworks that account for both return and risk dimensions. Benchmark comparison analysis reveals varying success rates for active management across categories, informing the active versus passive investment decision.

Risk and expense analysis demonstrates significant variation in risk profiles and cost structures across the mutual fund universe. The wide range of expense ratios observed within categories suggests that cost-conscious investors can identify value opportunities by comparing fees across similar products. Risk grade distribution confirms the availability of options spanning the full risk spectrum, supporting portfolio construction for diverse investor profiles.

NAV and portfolio analysis reveals the cumulative impact of investment decisions on fund valuations and the importance of sector diversification in managing concentration risk. SIP behaviour analysis confirms the democratisation of mutual fund investing through small, regular contributions, while geographic analysis identifies regional patterns that can guide distribution and marketing strategies.

Collectively, these findings establish a comprehensive understanding of the mutual fund landscape and provide the empirical foundation for the advanced analytics, risk metrics, and recommendation system developed in subsequent phases of this project.

6. Performance Analysis

Performance analysis in the context of mutual fund evaluation extends far beyond simple return calculations. While nominal returns provide a straightforward measure of how much an investment has grown over a specific period, they fail to capture the critical dimension of risk. Two funds may produce identical returns while exposing investors to vastly different levels of volatility and potential loss. Comprehensive performance analysis therefore requires the application of risk-adjusted metrics that evaluate returns in the context of the uncertainty and variability experienced during the investment period. This section details four fundamental performance metrics employed in this project: the Sharpe Ratio, Alpha, Beta, and the Sortino Ratio.

6.1 Sharpe Ratio

The Sharpe Ratio, developed by Nobel laureate William F. Sharpe, is perhaps the most widely recognised measure of risk-adjusted performance in finance. It quantifies the excess return generated per unit of total risk, where total risk is measured by the standard deviation of returns. The formula for the Sharpe Ratio is expressed as:

$$S = \frac{R_p - R_f}{\sigma_p} \quad (1)$$

where R_p represents the portfolio return, R_f denotes the risk-free rate of return (typically proxied by the yield on government securities), and σ_p signifies the standard deviation of portfolio returns. The Sharpe Ratio enables direct comparison between funds with different risk profiles by standardising returns by volatility. A higher Sharpe Ratio indicates superior risk-adjusted performance, meaning the fund generated more return for each unit of risk assumed. When evaluating mutual funds, investors should prefer schemes with higher Sharpe Ratios, as these funds have historically delivered better compensation for the volatility endured. A negative Sharpe Ratio indicates that the fund underperformed the risk-free rate, suggesting investors would have been better off holding risk-free assets.

6.2 Alpha

Alpha measures the excess return of a mutual fund relative to the return predicted by the Capital Asset Pricing Model (CAPM) given the fund's Beta and the market's performance. It represents the value added (or destroyed) by the fund manager's active investment decisions beyond what would be expected from market movements alone. Alpha is calculated as:

$$\alpha = R_p - [R_f + \beta(R_m - R_f)] \quad (2)$$

where R_m represents the market return and β is the fund's Beta coefficient. A positive Alpha indicates that the fund has outperformed its risk-adjusted benchmark expectation, suggesting skilled management, effective strategy execution, or favourable stock selection. Conversely, a negative Alpha indicates underperformance relative to the risk-adjusted expectation, raising questions about the value of active management after accounting for fees. Alpha is particularly important for investors paying active

management fees, as it directly addresses whether the manager has generated sufficient excess return to justify the cost premium over passive alternatives.

6.3 Beta

Beta quantifies the sensitivity of a mutual fund's returns to movements in the broader market. It measures systematic risk, which is the portion of total risk that cannot be eliminated through diversification because it stems from market-wide factors affecting all securities. The Beta coefficient is computed as:

$$\beta = \frac{\text{Cov}(R_p, R_m)}{\text{Var}(R_m)} \quad (3)$$

A Beta of one indicates that the fund's returns move in line with the market. A Beta greater than one signals higher volatility than the market, meaning the fund tends to amplify market movements both on the upside and downside. For example, a fund with a Beta of 1.2 would be expected to rise 12% when the market rises 10% and fall 12% when the market falls 10%. A Beta less than one indicates lower volatility than the market, suggesting a more conservative investment approach. Beta is essential for portfolio construction and risk management, as it helps investors understand how a fund will likely behave under different market scenarios and how it will contribute to the overall risk profile of a diversified portfolio.

6.4 Sortino Ratio

The Sortino Ratio is a refinement of the Sharpe Ratio that addresses a conceptual limitation in how risk is measured. While the Sharpe Ratio penalises both upside and downside volatility equally, the Sortino Ratio focuses exclusively on downside deviation, recognising that investors are primarily concerned with returns falling below a target or threshold level rather than volatility in either direction. The Sortino Ratio is calculated as:

$$\text{Sortino} = \frac{R_p - R_f}{\sigma_d} \quad (4)$$

where σ_d represents the standard deviation of negative returns (downside deviation). By isolating downside risk, the Sortino Ratio provides a more nuanced assessment of a fund's risk-return profile, particularly for strategies with asymmetric return distributions. Funds employing hedging techniques or options strategies may exhibit favourable Sortino Ratios relative to their Sharpe Ratios if they successfully limit downside volatility while preserving upside potential. For investors focused on capital preservation, the Sortino Ratio offers a more relevant risk metric than the Sharpe Ratio, as it directly addresses the risk of loss rather than total variability.

Together, these four metrics provide a comprehensive framework for mutual fund performance evaluation. The Sharpe Ratio offers a broad risk-adjusted performance measure, Alpha assesses manager skill, Beta characterises market sensitivity, and the Sortino Ratio refines risk assessment to focus on downside exposure. Investors utilising all four metrics in combination can develop a more complete understanding of a fund's historical performance characteristics and make more informed investment decisions aligned with their risk tolerance and return objectives.

7. Advanced Analytics and Risk Metrics

Beyond traditional performance metrics, this project implements several advanced analytical techniques that provide deeper insights into risk characteristics, investor behaviour patterns, and portfolio concentration. These methods address questions that conventional performance analysis cannot fully answer, such as the potential magnitude of losses under adverse conditions, the behavioural tendencies of different investor cohorts, and the degree of diversification within fund portfolios. This section describes each advanced analytical component, its methodology, and its practical implications for investment decision-making.

7.1 Historical Value at Risk (VaR)

Value at Risk (VaR) is a statistical measure that quantifies the maximum potential loss of an investment portfolio over a specified time horizon at a given confidence level. The Historical VaR method employed in this project uses actual historical return data to estimate potential losses, making no assumptions about the distribution of returns. This non-parametric approach is particularly valuable when return distributions exhibit fat tails or skewness that would violate the normality assumptions underlying parametric VaR models.

The methodology for computing Historical VaR at the 95% confidence level involves the following steps. First, the historical daily returns of the mutual fund scheme are computed from the NAV time series data. Second, these returns are sorted in ascending order from the most negative to the most positive. Third, the 5th percentile of the return distribution is identified, representing the threshold below which only 5% of historical observations have fallen. This 5th percentile value represents the VaR, indicating the loss level that would be exceeded with only 5% probability based on historical experience.

Interpretation of VaR requires careful consideration of its limitations and assumptions. A 95% Historical VaR of negative two percent implies that, based on historical data, there is a 5% chance of losing more than two percent of the portfolio value over a single day. However, VaR does not indicate how much worse than two percent the loss could be in those rare adverse scenarios, nor does it guarantee that future losses will conform to historical patterns. Despite these limitations, VaR remains a valuable risk communication tool, providing a single, intuitive number that captures the scale of potential downside exposure and facilitates comparison across different funds and portfolios.

7.2 Conditional Value at Risk (CVaR)

Conditional Value at Risk (CVaR), also known as Expected Shortfall, addresses one of the primary limitations of VaR by quantifying the expected magnitude of losses in scenarios where the VaR threshold is breached. While VaR answers the question "How bad can things get with a certain probability?", CVaR answers the follow-up question "If things get that bad, how much can we expect to lose on average?" This tail risk measurement provides critical information for stress testing and risk management, as it focuses on the most adverse scenarios rather than the boundary of normal conditions.

CVaR is computed as the average of all historical returns that fall below the VaR threshold. Mathematically, for a 95% confidence level, CVaR represents the mean of the worst 5% of historical

returns. This computation ensures that CVaR is always at least as large as VaR in magnitude, with the difference between VaR and CVaR indicating the severity of tail risk. A large gap between VaR and CVaR suggests significant tail risk, meaning that extreme losses can be substantially worse than the VaR threshold would suggest.

The importance of CVaR in financial decision-making cannot be overstated, particularly in the aftermath of market crises where extreme events occur more frequently than standard statistical models would predict. For mutual fund investors, CVaR provides a more conservative and comprehensive view of downside risk than VaR alone. Funds with seemingly acceptable VaR levels but high CVaR values may expose investors to catastrophic losses during market stress periods. By incorporating both VaR and CVaR into the risk assessment framework, this project enables a more robust evaluation of mutual fund risk profiles that accounts for both the probability and the severity of adverse outcomes.

7.3 Rolling 90-Day Sharpe Ratio

The Rolling 90-Day Sharpe Ratio provides a dynamic view of risk-adjusted performance over time, complementing the single-period Sharpe Ratio with a time series that reveals how a fund's risk-return efficiency has evolved. By computing the Sharpe Ratio over a rolling 90-day window, this analysis captures recent trends in performance quality and identifies periods of improving or deteriorating risk-adjusted returns.

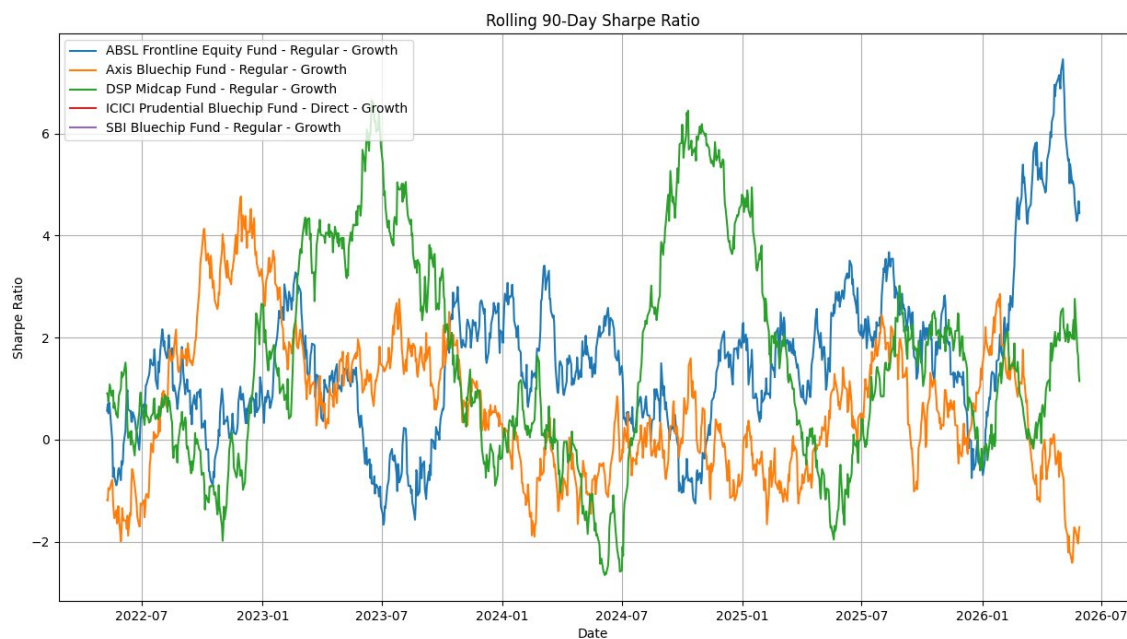


Figure 19: Rolling 90-Day Sharpe Ratio Trend (rolling_sharpe_chart.png)

Figure 19 presents the Rolling 90-Day Sharpe Ratio for selected mutual fund schemes. The trend line reveals important patterns in risk-adjusted performance stability. A consistently positive rolling Sharpe Ratio indicates sustained risk-adjusted outperformance, while oscillations between positive and negative values suggest volatile performance quality. Sharp declines in the rolling Sharpe Ratio may signal deteriorating performance, increased volatility, or both, warranting closer investigation. Comparing rolling Sharpe trends across multiple funds reveals which schemes have maintained consistent risk-

adjusted performance and which have exhibited greater variability. For portfolio construction, funds with stable, positive rolling Sharpe Ratios are generally preferred over those with high average Sharpe Ratios but significant time variation, as consistency reduces uncertainty in future performance expectations.

7.4 Investor Cohort Analysis

Investor cohort analysis is a behavioural analytics technique that groups investors based on shared characteristics or experiences and tracks their behaviour over time. In this project, cohorts were formed based on the month of first investment (entry cohort), enabling the analysis of how investment behaviour varies between groups that entered the market at different times and under different market conditions.

The cohort formation process involves grouping investors by their entry month and then tracking metrics such as retention rate, additional investment frequency, redemption patterns, and SIP continuation rates for each cohort across subsequent months. This longitudinal analysis reveals how market conditions at entry influence long-term investor behaviour. For example, investors who entered during bull markets may exhibit different retention characteristics compared to those who entered during bear markets or periods of high volatility.

Major observations from the cohort analysis typically include the identification of entry timing effects on investor longevity, the discovery of cohort-specific behavioural patterns, and the quantification of investor lifetime value by entry period. These insights have practical applications in investor onboarding programmes, as understanding the characteristics of high-retention cohorts enables the development of targeted interventions for new investors. Additionally, cohort analysis helps identify periods of market stress that had lasting impacts on investor behaviour, informing the design of communication and support strategies during future downturns.

7.5 SIP Continuity Analysis

SIP continuity analysis focuses on identifying investors at risk of discontinuing their systematic investment plans. Given that SIPs represent the backbone of retail mutual fund participation in India, understanding and predicting discontinuation risk is critical for both investor welfare and industry stability.

The analysis employs multiple indicators to identify at-risk investors, including declining investment amounts, increased redemption activity, missed SIP instalments, and changes in investor communication patterns. By combining these signals, a risk score is computed for each SIP investor, categorising them into risk segments from low to high discontinuation probability.

The importance of SIP continuity extends beyond individual investor outcomes to industry-level concerns. High discontinuation rates during market downturns can force fund managers to liquidate holdings at unfavourable prices, potentially affecting remaining investors. Furthermore, the pattern of discontinuing during downturns and restarting during recoveries represents a classic buy-high, sell-low behaviour that undermines wealth creation. By identifying at-risk investors proactively, Asset Management Companies can implement retention interventions such as educational communications, portfolio review offers, and flexible payment options that help investors maintain their systematic investment discipline through volatile periods.

7.6 Mutual Fund Recommendation System

The mutual fund recommendation system developed in this project translates complex analytical findings into actionable investment guidance by matching investor risk profiles with suitable fund categories. The system operates on a risk-based segmentation framework that categorises investors into distinct risk appetite categories.

Investors are classified into risk categories based on their stated risk tolerance, investment horizon, financial goals, and demographic characteristics. The typical categorisation includes Conservative investors prioritising capital preservation, Moderate investors seeking balanced growth and stability,

Aggressive investors willing to accept higher volatility for potentially superior returns, and Very Aggressive investors targeting maximum growth with full acceptance of significant downside risk.

The recommendation logic selects funds within each risk category based on their Sharpe Ratio rankings, ensuring that recommended funds have demonstrated superior risk-adjusted performance within their peer group. Additional filtering criteria include minimum AUM thresholds to ensure liquidity, maximum expense ratio limits to protect investor returns, and performance consistency requirements to avoid recommending funds with erratic return patterns. The system generates a ranked list of recommended schemes for each risk category, providing investors with a curated selection of funds that align with both their risk capacity and return objectives.

7.7 Sector HHI Concentration Analysis

The Herfindahl-Hirschman Index (HHI) is a widely used measure of market concentration that has been adapted in this project to assess portfolio concentration at the sector level. Portfolio concentration represents an important risk dimension, as overexposure to a single sector can amplify losses during industry-specific downturns.

The HHI is computed as the sum of the squared percentage allocations across all sectors in the portfolio:

$$HHI = \sum_{i=1}^n s_i^2 \quad (5)$$

where s_i represents the percentage allocation to sector i and n is the total number of sectors. The HHI ranges from a minimum value achieved with equal allocation across all sectors (indicating maximum diversification) to a maximum of 10,000 for a portfolio concentrated entirely in a single sector. Higher HHI values indicate greater concentration risk, while lower values suggest more diversified portfolios.

The implications of HHI analysis for investors are substantial. Funds with very high sector HHI values expose investors to concentrated risk that may not be apparent from overall portfolio statistics. A technology-heavy portfolio, for example, may exhibit attractive historical returns during periods of tech sector outperformance but suffer disproportionately during tech downturns. Comparing HHI values across funds within the same category enables investors to identify those managers taking concentrated bets versus those pursuing more diversified approaches. The optimal level of concentration depends on investor preferences, with some investors favouring focused portfolios where manager conviction is high, while others prioritise broad diversification to minimise idiosyncratic risk.

8. Power BI Dashboard Development

The Power BI Dashboard Development phase represents the culmination of the analytical work, transforming complex data findings into intuitive, interactive visualisations that enable stakeholders to explore mutual fund data dynamically. Power BI was selected as the business intelligence platform due to its robust data modelling capabilities, extensive visualisation library, interactive filtering features, and seamless integration with the Python-based analytical pipeline. This section describes the dashboard design philosophy, technical implementation, and the insights provided by each dashboard page.

8.1 Dashboard Design Philosophy

The dashboard design followed a user-centric approach, prioritising clarity, accessibility, and actionable insight generation. The primary objective was to create a tool that serves multiple stakeholder needs, from retail investors seeking simplified fund comparisons to financial advisors requiring detailed analytical capabilities. Key Performance Indicators (KPIs) were carefully selected to highlight the most critical metrics at a glance, while underlying detailed data remains accessible through drill-through functionality.

The visual design employs a consistent colour scheme and layout pattern across all pages to ensure navigational familiarity. Data visualisations were chosen based on their effectiveness for the specific data type being presented, with line charts for trends, bar charts for comparisons, heatmaps for pattern identification, and cards for single-value metrics. Slicers and filters are prominently positioned to enable dynamic data exploration without overwhelming the primary visual content.

8.2 Technical Implementation

The technical implementation involved importing the processed datasets from the ETL pipeline into Power BI's data model. Relationships were established between fact tables (NAV history, transactions, performance metrics) and dimension tables (fund master, date table, category mappings) using primary and foreign key joins. Data types were verified, and calculated columns and measures were created using Data Analysis Expressions (DAX) to support dynamic computations such as rolling averages, year-to-date returns, and category rankings.

Filters and slicers were configured to enable multi-dimensional data exploration. Users can filter data by date range, fund category, AMC, risk grade, and AUM size, with all visualisations on a page responding synchronously to filter selections. Drill-through functionality allows users to navigate from summary-level views to detailed transaction-level data, enabling both high-level pattern recognition and granular investigation when anomalies or areas of interest are identified.

Theme customisation was applied to ensure professional presentation with optimal readability. The colour palette was selected to provide sufficient contrast for accessibility while maintaining visual appeal. Conditional formatting was applied to highlight exceptional values, such as funds with Sharpe Ratios significantly above category averages or expense ratios exceeding predefined thresholds.

8.3 Dashboard Page 1: Executive Overview

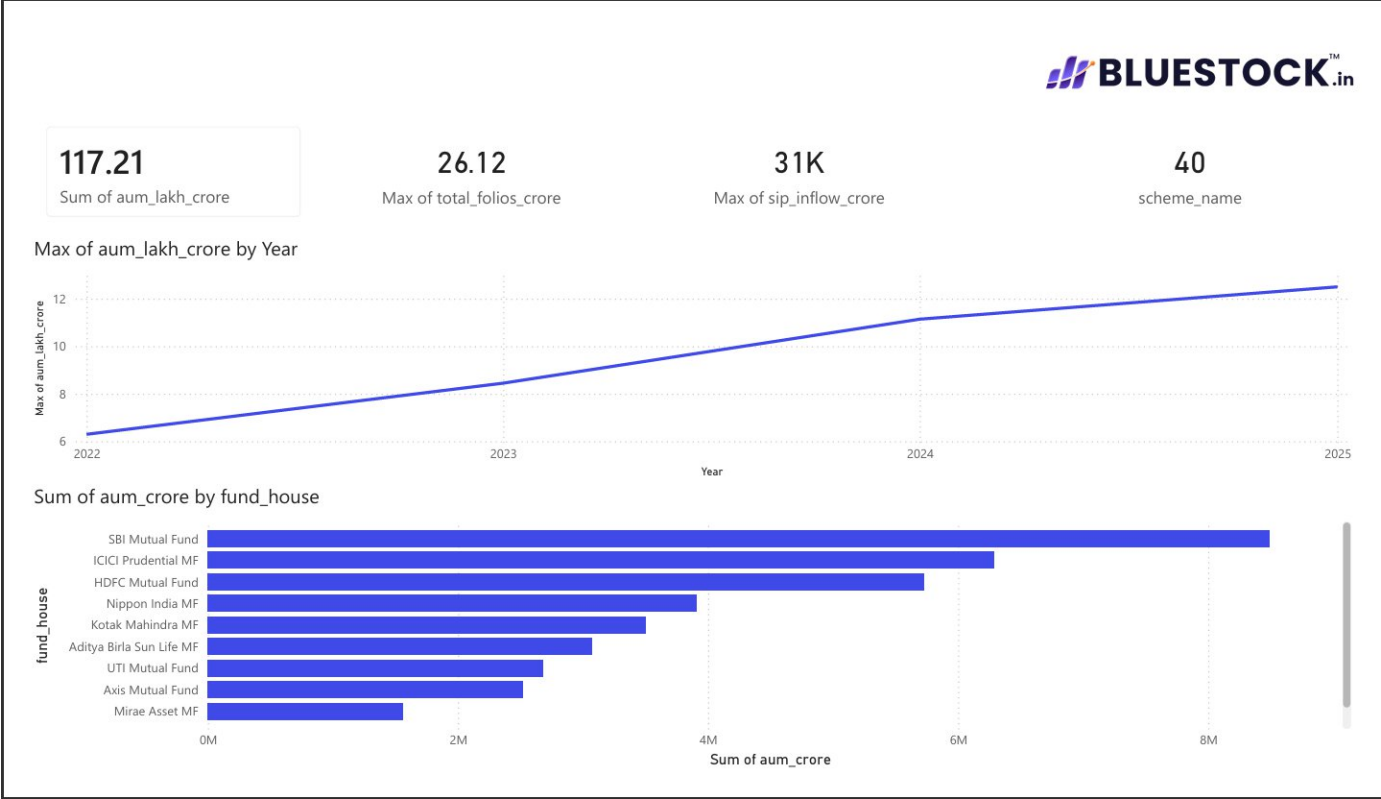


Figure 20: Power BI Dashboard - Executive Overview Page

Figure 20 presents the Executive Overview page, designed as the primary landing view providing a high-level summary of the mutual fund landscape. This page features KPI cards displaying total AUM, total folio count, average category return, and industry-wide SIP inflows for the most recent period. A trend line chart shows AUM growth over time, while a bar chart ranks the top ten AMCs by market share. A category-wise breakdown donut chart illustrates the distribution of AUM across fund types. The purpose of this page is to provide senior stakeholders and decision-makers with an immediate understanding of industry scale, growth trajectory, and competitive dynamics. The insights generated include identification of market leaders, recognition of growth trends, and awareness of category-level shifts in investor preferences.

8.4 Dashboard Page 2: Fund Performance Explorer

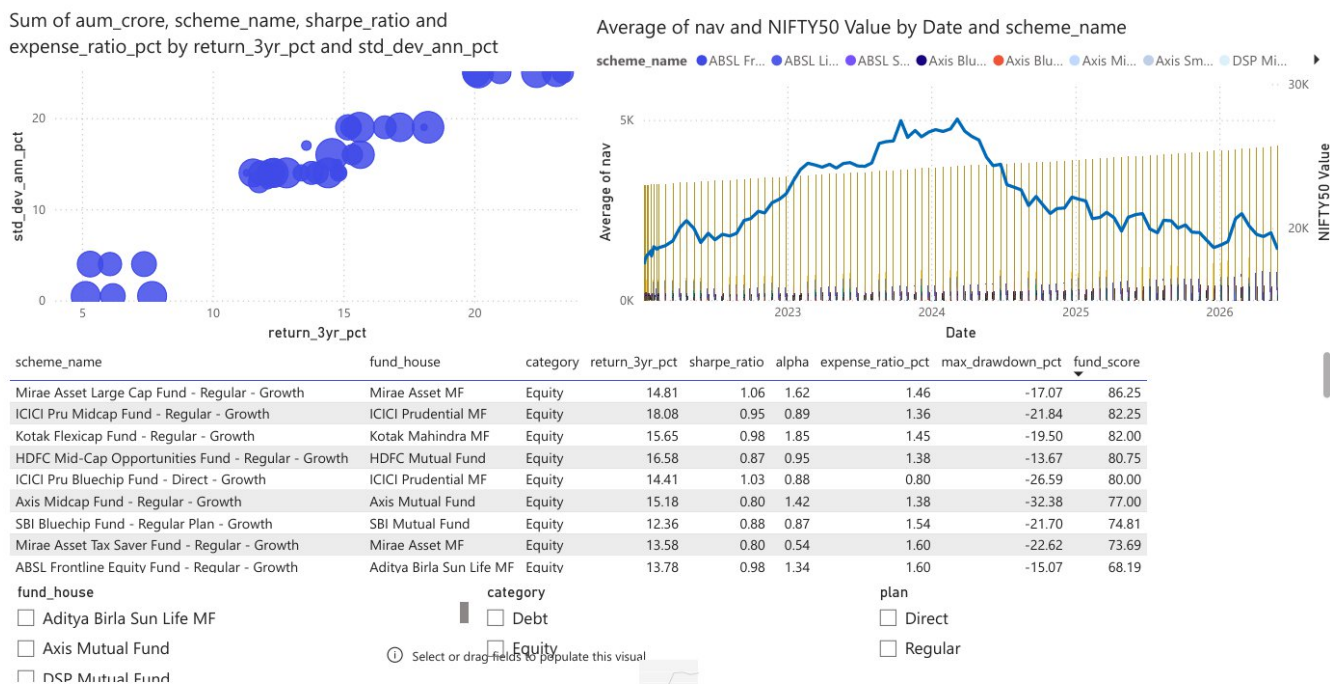


Figure 21: Power BI Dashboard - Fund Performance Explorer Page

Figure 21 displays the Fund Performance Explorer page, which enables detailed analysis of individual scheme performance across multiple metrics. This page includes a scatter plot mapping funds by return versus risk, with bubble size representing AUM, enabling rapid identification of efficient frontier funds. A matrix table lists key performance metrics including returns, Sharpe Ratio, Alpha, Beta, and Sortino Ratio for selected funds, sortable by any column. Bar charts compare category average returns and highlight top performers. Filter slicers allow users to narrow the universe by category, risk grade, AMC, and AUM range. This page serves financial advisors and sophisticated investors conducting fund due diligence. The insights provided include identification of funds with optimal risk-return profiles, comparison of peer group performance, and detection of outliers warranting further investigation.

8.5 Dashboard Page 3: Investor Analytics

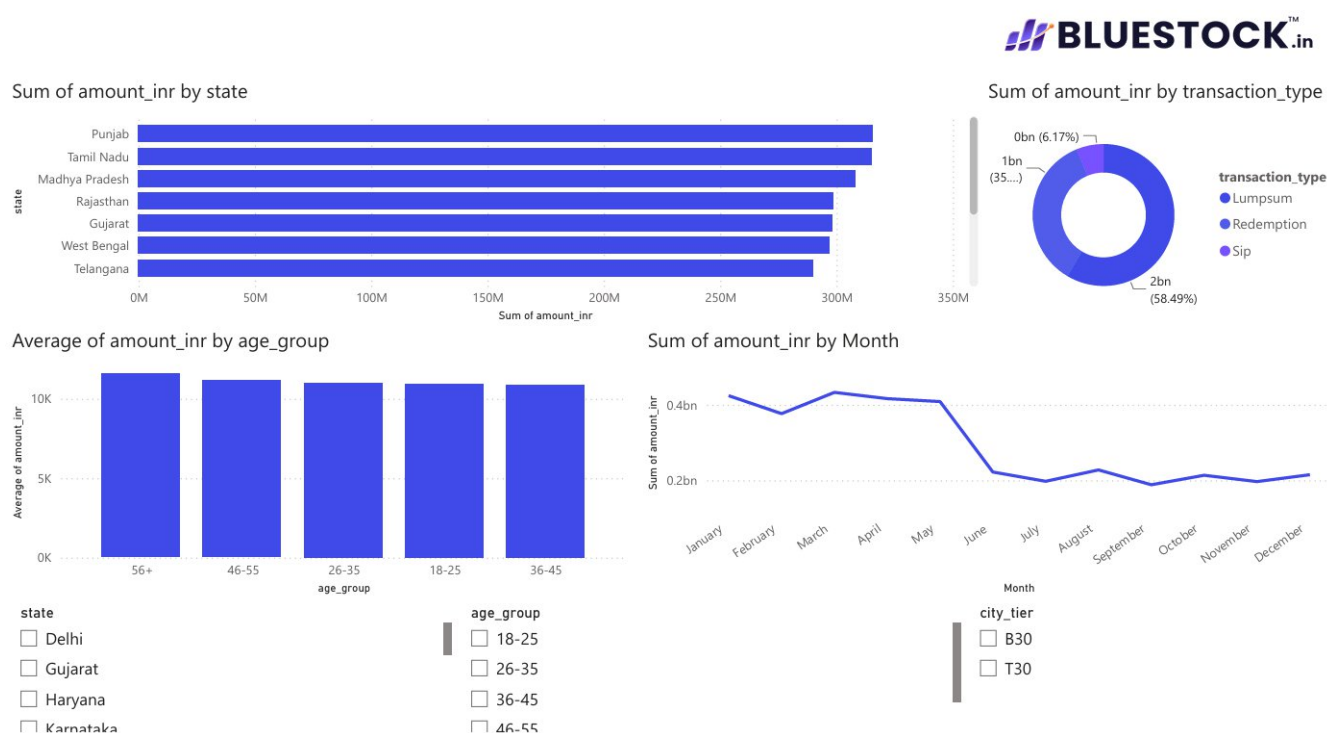


Figure 23: Power BI Dashboard - Investor Analytics Page

Figure 23 shows the Investor Analytics page, which focuses on understanding investor behaviour and demographics. This page features demographic distribution charts for age, gender, and city tier, enabling profiling of the investor base. A funnel chart illustrates the investor journey from initial registration through first transaction to SIP enrollment. A line chart tracks SIP retention rates by cohort entry month, highlighting discontinuation patterns. A geographic map visualises state-wise investment density. This page is particularly valuable for product managers, marketing teams, and distribution strategists seeking to understand and optimise investor engagement. Key insights include identification of high-value investor segments, recognition of geographic expansion opportunities, and detection of cohorts with elevated discontinuation risk requiring intervention.

8.6 Dashboard Page 4: Risk and Portfolio Analysis

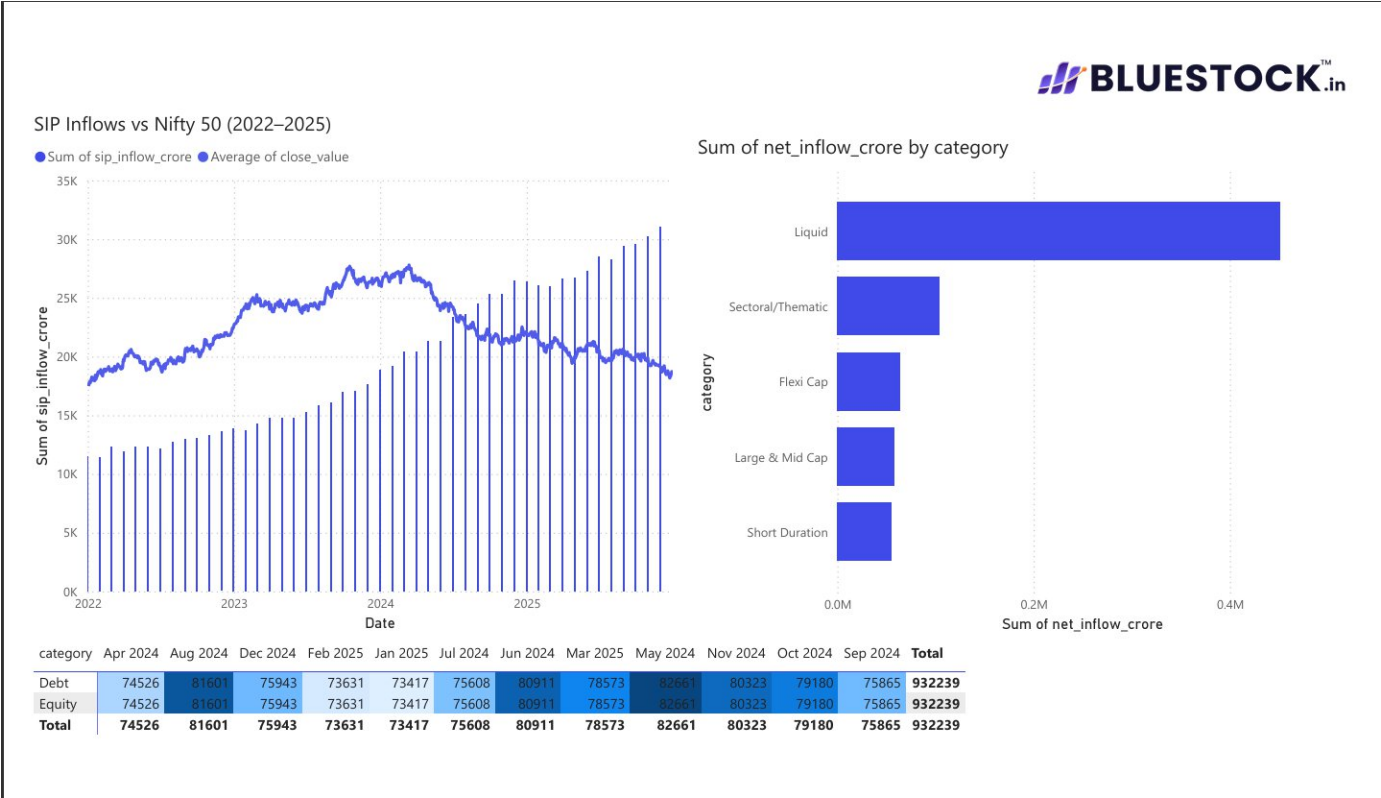


Figure 23: Power BI Dashboard - Risk and Portfolio Analysis Page

Figure 23 presents the Risk and Portfolio Analysis page, providing tools for deep risk assessment and portfolio composition examination. This page includes a risk grade distribution chart showing the proportion of funds across risk categories. A bar chart displays VaR and CVaR values by fund, enabling comparison of downside risk exposure. The sector HHI gauge visual indicates concentration risk levels for selected funds. A treemap shows sector allocation breakdowns, revealing portfolio composition at a glance. A correlation heatmap displays return correlations between selected funds, supporting diversification analysis. This page serves risk managers, portfolio analysts, and compliance teams responsible for monitoring risk exposures. The insights generated include identification of funds with elevated tail risk, assessment of portfolio concentration, and evaluation of diversification benefits from combining specific funds.

9. Key Findings

The comprehensive analytical framework developed in this project has generated several significant findings that contribute to the understanding of mutual fund performance, investor behaviour, and risk characteristics. These findings represent the synthesis of insights derived from the ETL process,

exploratory data analysis, performance evaluation, advanced analytics, and dashboard visualisation components of the project.

Finding 1: Risk-Adjusted Performance Varies Significantly Within Categories. Analysis of Sharpe Ratios across funds within the same category reveals substantial variation in risk-adjusted performance, with top-quartile funds delivering Sharpe Ratios more than double those of bottom-quartile funds. This finding challenges the common investor assumption that funds within the same category are broadly similar. In reality, fund selection within a category can have a profound impact on the risk-return experience. The variation suggests that manager skill, process discipline, and cost efficiency create meaningful differentiation even among funds with ostensibly similar mandates. Investors should therefore conduct rigorous due diligence within categories rather than selecting funds based solely on recent return performance or brand recognition.

Finding 2: SIP Continuity is Strongly Correlated with Entry Timing and Market Conditions. Cohort analysis reveals that investors who initiated SIPs during market corrections or periods of elevated volatility demonstrate higher long-term continuation rates compared to those who entered during sustained bull markets. This counterintuitive finding suggests that investors who experience early volatility develop more realistic expectations about market behaviour and the importance of disciplined investing. Conversely, investors entering during strong market periods may develop unrealistic return expectations that lead to disappointment and discontinuation when normal market volatility resumes. This insight has important implications for investor education programmes, suggesting that preparing investors for volatility may be more effective than emphasising historical returns during onboarding.

Finding 3: Expense Ratios Show Limited Correlation with Performance. The analysis of expense ratios relative to risk-adjusted performance metrics reveals no consistent positive relationship between higher fees and superior outcomes. In several categories, lower-cost funds actually delivered better risk-adjusted returns on average, suggesting that cost-conscious fund selection can enhance investor outcomes. This finding supports the growing global trend towards cost transparency and fee compression in asset management. Investors and advisors should critically evaluate whether higher expense ratios are justified by demonstrably superior performance, portfolio construction complexity, or service quality, rather than assuming that higher costs equate to better management.

Finding 4: Portfolio Concentration Risk is Underappreciated by Retail Investors. HHI analysis of mutual fund portfolios reveals that a significant proportion of schemes exhibit high sector concentration, with some equity funds allocating over 40% of their portfolio to a single sector. Despite this concentration risk, investor flows into these funds remain strong, suggesting that retail investors may not fully understand or account for sector-specific risk in their fund selection process. The correlation between concentrated portfolios and periods of strong recent performance indicates that investors may be inadvertently chasing returns while underestimating the vulnerability of concentrated strategies to sector-specific downturns. Enhanced disclosure of concentration metrics and investor education about diversification benefits could help address this gap.

Finding 5: Benchmark Outperformance is Concentrated Among a Small Subset of Funds. Analysis of fund performance relative to benchmarks reveals that consistent outperformance is achieved by a relatively small proportion of active managers. The majority of funds either underperform their

benchmarks after accounting for expenses or deliver inconsistent results that do not justify active management fees. However, the subset of funds that do demonstrate persistent outperformance tends to maintain their edge over multi-year periods, suggesting that manager skill, where it exists, is somewhat persistent. This finding supports a selective approach to active management, where investors focus due diligence efforts on identifying funds with demonstrated process discipline, consistent philosophy, and long-term track records rather than broadly allocating across all active funds in a category.

10. Limitations

While this project has achieved its stated objectives and generated valuable insights, it is important to acknowledge several limitations that constrain the scope and generalisability of the findings. Understanding these limitations provides context for interpreting the results and identifies areas for improvement in future work.

Static Datasets. The analysis was conducted on static datasets representing a snapshot of mutual fund information at a specific point in time. Unlike real-time data feeds that continuously update with new NAV values, transaction records, and portfolio changes, the static datasets used in this project do not reflect the most current market conditions. This limitation means that the specific performance rankings, risk metrics, and recommendations generated may have changed since the data was captured. For production deployment, the analytical framework would need to be adapted to process streaming or frequently updated data to ensure that insights remain current and actionable.

Lack of Real-Time Integration. The ETL pipeline and analytical processes were designed for batch processing of historical data rather than real-time or near-real-time data integration. In a live operational environment, mutual fund analytics platforms must handle continuous data ingestion, incremental updates, and low-latency query responses. The batch-oriented approach employed in this project, while appropriate for a capstone demonstration, would require significant architectural modification to support production-scale real-time analytics. This includes considerations for data pipeline orchestration, change data capture, and streaming analytics infrastructure.

Simplified Recommendation Logic. The mutual fund recommendation system implemented in this project uses a rules-based approach that matches investor risk profiles with funds ranking highly on the Sharpe Ratio within the corresponding risk category. While this approach provides a logical starting point, it represents a simplified model compared to the sophisticated recommendation engines deployed by leading financial platforms. Advanced recommendation systems incorporate collaborative filtering, content-based filtering, natural language processing of fund documents, and machine learning models that capture complex nonlinear relationships between investor characteristics and fund suitability. The simplified logic may not capture nuanced preferences or identify novel recommendations that diverge from conventional risk-based matching.

Limited Historical Scope. The historical depth of the available data constrains the ability to analyse performance across complete market cycles. Mutual fund evaluation ideally requires data spanning multiple bull and bear markets, economic cycles, and interest rate regimes to assess strategy robustness under diverse conditions. The datasets used in this project, while substantial, may not encompass periods of severe market stress or structural market changes that test the resilience of investment strategies. Long-

term performance persistence, risk metric stability, and the reliability of VaR and CVaR estimates all benefit from extended historical observation periods.

11. Recommendations

Based on the findings, limitations, and insights generated throughout this project, the following recommendations are offered for enhancing the mutual fund analytics framework and extending its practical applicability in investment decision-making contexts.

Real-Time Data Integration. The most impactful enhancement would be the integration of real-time data feeds from AMFI, stock exchanges, and market data providers. Real-time NAV updates, transaction processing, and portfolio rebalancing information would transform the current static analytical framework into a dynamic decision support system. This enhancement would require the implementation of streaming data architecture using technologies such as Apache Kafka for data ingestion, Apache Spark for stream processing, and real-time database solutions for low-latency querying. The ability to monitor portfolio risk, track performance, and receive alerts on an intraday basis would significantly enhance the utility of the platform for professional investors and fund managers.

Machine Learning Enhancements. The recommendation system and predictive analytics components would benefit substantially from the incorporation of machine learning algorithms. Supervised learning models such as gradient boosting machines and random forests could improve fund return predictions by capturing nonlinear relationships between fund characteristics and future performance. Unsupervised clustering algorithms could identify natural groupings of funds with similar behaviour patterns, enabling more nuanced peer group comparisons. Reinforcement learning approaches could optimise portfolio allocation strategies by learning from historical market dynamics. Natural language processing could analyse fund manager commentary, regulatory filings, and news sentiment to generate qualitative insights complementing the quantitative metrics.

Automated Reporting. The development of automated reporting capabilities would streamline the distribution of analytical insights to stakeholders. Scheduled report generation, exception-based alerting, and self-service report customisation would reduce manual effort and ensure that decision-makers receive timely, relevant information. Integration with email distribution, collaboration platforms, and document management systems would facilitate seamless information sharing across organisations. Automated commentary generation using natural language generation techniques could produce narrative explanations of performance changes, risk metric movements, and market developments, making analytical outputs more accessible to non-technical stakeholders.

Advanced Portfolio Optimisation. Extending the analytical framework to include portfolio-level optimisation would enable users to construct efficient portfolios rather than simply selecting individual funds. Mean-variance optimisation, Black-Litterman models, and risk parity approaches could be implemented to generate optimal fund allocations based on investor objectives, constraints, and market views. Monte Carlo simulation could project portfolio outcomes under various scenarios, helping investors understand the range of potential future results. Tax-aware optimisation could account for the specific tax implications of different fund types and transaction strategies in the Indian regulatory context, enhancing after-tax returns.

12. Future Scope

The foundation established by this project opens several promising avenues for future development and research. These extensions would build upon the analytical infrastructure, methodologies, and insights developed during the capstone programme, progressively enhancing the sophistication and practical value of the mutual fund analytics platform.

Robo-Advisory Systems. The recommendation system developed in this project could serve as the analytical core of a comprehensive robo-advisory platform. Robo-advisors automate the investment advice process, using algorithms to assess investor profiles, recommend suitable portfolios, execute trades, and rebalance holdings with minimal human intervention. Integration with account aggregation services, know-your-customer verification systems, and payment gateways would enable end-to-end digital wealth management. The scalability of robo-advisory models makes them particularly suitable for serving the mass affluent segment and Tier 2 and Tier 3 city populations that may be underserved by traditional financial advisory channels.

Predictive Analytics. Future development could incorporate predictive models that forecast fund performance, investor behaviour, and market trends. Time series forecasting models such as ARIMA, Prophet, and LSTM neural networks could predict NAV movements, AUM flows, and category rotation patterns. Survival analysis could model investor lifecycle events such as SIP discontinuation or full redemption. These predictive capabilities would enable proactive rather than reactive decision-making, allowing fund managers, distributors, and investors to anticipate developments and position accordingly.

Mobile Dashboards. The development of mobile-optimised dashboard applications would extend the reach of mutual fund analytics to smartphone users. Mobile dashboards require thoughtful design adaptation to accommodate smaller screens, touch interactions, and variable network conditions. Progressive web application technology or native mobile development could deliver the core dashboard functionality to investors and advisors who primarily access information through mobile devices. Push notifications for significant portfolio events, market alerts, and personalised insights would enhance engagement and timeliness.

Real-Time Monitoring. A real-time monitoring system would provide continuous surveillance of portfolio risk metrics, market conditions, and regulatory compliance parameters. Such a system could trigger automated alerts when risk thresholds are breached, when significant market events occur, or when portfolio characteristics drift from target allocations. Integration with execution management systems could enable automated rebalancing, stop-loss implementation, and tax-loss harvesting. Real-time monitoring is particularly valuable for institutional investors, portfolio management services, and regulatory compliance functions that require immediate awareness of changing conditions.

13. Conclusion

The Bluestock Mutual Fund Analytics Capstone Project has successfully achieved all nine objectives defined at the project's inception, delivering a comprehensive analytical framework that addresses the challenges identified in the problem statement. The project demonstrates the practical application of data engineering, statistical analysis, financial modelling, and business intelligence techniques to the domain of mutual fund evaluation and investment decision support.

The ETL pipeline designed and implemented during this project established a robust data foundation, processing nine distinct datasets through systematic extraction, transformation, and loading procedures. The pipeline's handling of missing values, deduplication, standardisation, feature engineering, and dataset integration ensures that downstream analyses operate on high-quality, consistent data. This data infrastructure represents a reusable asset that can accommodate new data sources and updated datasets as the analytical framework evolves.

The analytical techniques implemented span the full spectrum from descriptive to predictive analytics. Exploratory data analysis uncovered meaningful patterns in investor demographics, industry trends, fund performance, risk profiles, and expense structures. Performance analysis using the Sharpe Ratio, Alpha, Beta, and Sortino Ratio provided standardised frameworks for risk-adjusted fund comparison. Advanced analytics including Historical VaR, CVaR, rolling performance metrics, cohort analysis, SIP continuity assessment, and HHI concentration measurement addressed sophisticated questions about tail risk, investor behaviour, and portfolio composition. The recommendation system translated these analytical findings into practical guidance aligned with investor risk profiles.

The business value generated by this project extends across multiple stakeholder groups. Retail investors gain access to data-driven fund selection tools that move beyond simplistic return comparisons. Financial advisors can leverage the comprehensive metrics and interactive dashboards to enhance client communication and portfolio construction. Asset Management Companies can utilise the behavioural analytics to improve product design, distribution strategy, and investor retention. The Power BI dashboards consolidate these capabilities into an accessible interface that democratises access to institutional-quality analytics.

The significance of mutual fund analytics in the contemporary financial landscape cannot be overstated. As the Indian mutual fund industry continues its trajectory of growth and investor participation broadens across demographic and geographic dimensions, the need for sophisticated analytical tools becomes increasingly acute. This project contributes to that need by providing a replicable, extensible framework that bridges the gap between raw data abundance and actionable investment intelligence. The methodologies, code, and insights developed during this internship will serve as a foundation for continued learning and professional development in the intersection of data science and financial services.

